Part VII

Exotics and Structured Products

These two chapters treat payoffs that are designed rather than listed. The first covers the exotic families and the replication arguments that price them; the second follows those payoffs into the retail market, where the same structure is sold as a note with a headline coupon. Read together they make one point: the exotic is priced by the dealer and bought by someone reading a different document.

34 Exotic Options
barriers, lookbacks, Asians, baskets, and the arguments that price them

35 Structured Products and Autocallables
the same payoffs wrapped as notes, and what the wrapper hides

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Chapter 34
Exotic Options

Five chapters have deferred material to this one. Almost everything in it trades over the counter, in the market Section 10.10 described; an exotic option is a dealer's product, and that fact shapes how it is priced, hedged, and sold.

Section 15.2 said every payoff whose value depends on S and t alone satisfies the Black–Scholes–Merton equation without modification, and that a path-dependent one adds a state variable. Section 18.6 said vanna and volga matter here. Section 18.13 said that where gamma diverges, dynamic hedging becomes impossible and a static hedge is the only answer. Section 19.12 said barrier prices depend on how the surface *evolves*, which no static surface specifies. Section 20.4 promised that a barrier priced on a tree whose nodes straddle the barrier converges erratically. Section 28.11 said the multi-asset family of options is developed here.

The unifying question is which exotics are genuinely harder than vanillas and which merely look it. The answer is not about payoff complexity. It is about whether the payoff can be replicated by instruments the market quotes, and Sections 34.5 and 34.13 give the test.

34.1 WHAT MAKES AN OPTION EXOTIC

Three features separate an exotic from a vanilla, with very different consequences. Path dependence turns on the route rather than the endpoint, so its notation comes first. Throughout, S_t is the underlying and K the strike, and the path statistics are

$$M_T = \max_{0\le t\le T} S_t, \qquad m_T = \min_{0\le t\le T} S_t, \qquad A_T = \frac{1}{T}\int_0^T S_t\,dt$$

the running maximum, the running minimum, and the average.

Path dependence. The payoff depends on the route, not only the destination. A **barrier option** is a vanilla multiplied by an indicator of whether the path has reached a barrier level H:

$$\text{down-and-out call} = (S_T-K)^+\mathbf{1}_{\{m_T>H\}}, \qquad \text{down-and-in call} = (S_T-K)^+\mathbf{1}_{\{m_T\le H\}} \tag{34.1}$$

with the up-variants replacing m_T by M_T. Written this way, in–out parity (knock-in plus knock-out equals the vanilla, equation (34.10) in Section 34.3) stops being a result and becomes an observation: the two indicators sum to one, so the two options sum to the vanilla.

A **lookback option** replaces the strike or the terminal price by an extreme of the path:

$$\begin{aligned} \text{floating call} &= S_T - m_T, & \text{floating put} &= M_T - S_T,\\ \text{fixed call} &= (M_T - K)^+, & \text{fixed put} &= (K - m_T)^+ \end{aligned} \tag{34.2}$$

The floating pair can never finish out of the money, which is most of the reason they are expensive. An **Asian option** replaces the terminal price by the average A_T in one of two places:

$$\text{average price} = (A_T - K)^+, \qquad \text{average strike} = (S_T - A_T)^+ \tag{34.3}$$

and a **cliquet** sums period returns $r_i = S_{t_i}/S_{t_{i-1}} - 1$ over reset dates $0 = t_0 < t_1 < \cdots < t_n = T$,

each floored at F and capped at C:

$$\text{cliquet} = \sum_{i=1}^{n} \min\big(\max(r_i,\ F),\ C\big) \tag{34.4}$$

The cap is inside the sum rather than outside it, and Section 34.12 is about what that does to the sign of the volatility exposure.

This is where the model risk lives, because a model calibrated to terminal distributions does not pin down paths. Five sections of this chapter go to the barrier alone for that reason (Sections 34.3 through 34.7), and Section 34.9 takes the lookback and the Asian together.

Multiple underlyings. Basket options pay on a weighted sum of several assets, **rainbow options** on the best or the worst of them, and **exchange options** on one asset against another; four payoffs matter:

$$\begin{aligned} \text{basket} &= \Big(\sum_i w_i S_i(T) - K\Big)^+, & \text{exchange} &= \big(S_1(T) - S_2(T)\big)^+, \\ \text{worst-of put} &= \big(K - \min_i S_i(T)\big)^+, & \text{best-of call} &= \big(\max_i S_i(T) - K\big)^+ \end{aligned} \tag{34.5}$$

Here $S_i(T)$ is the price of the ith asset at expiration and w_i the number of units of it the basket holds; the basket of Example 34.8 is $(S_1 + S_2)/2$, so $w_1 = w_2 = 1/2$. The new input is correlation, which Section 19.7 established is not observable in the way volatility is. It enters every one of these payoffs and is quoted in none of them, which is Section 34.10's subject and the reason the worst-of put in Section 34.11 behaves as it does.

Non-standard exercise or payoff. This group holds Bermudans, compounds, choosers, and the digital pair (options paying either one unit of cash or the asset itself if they finish in the money), and the last of these carries the identity that ties it to the Black–Scholes–Merton formula:

$$\begin{aligned} \text{cash-or-nothing} &= \mathbf{1}_{\{S_T > K\}}, & \text{asset-or-nothing} &= S_T\,\mathbf{1}_{\{S_T > K\}}, \\ (S_T - K)^+ &= S_T\,\mathbf{1}_{\{S_T > K\}} - K\,\mathbf{1}_{\{S_T > K\}} \end{aligned} \tag{34.6}$$

A vanilla call is one asset-or-nothing less K cash-or-nothings, which is Section 28.10's reading of $N(d_1)$ and $N(d_2)$ written as a payoff rather than as a price. These are usually the easy ones, because a single-asset payoff at a single date is a function of the terminal distribution, which the vanilla market already reveals.

The taxonomy sorts the chapter: the digitals and the simple chooser are priced almost model-independently, while the first two groups are not. A desk that treats "exotic" as one category prices some trades far too carefully and others not carefully enough.

34.2 OPTIONS THAT ARE BARELY EXOTIC

Several instruments carry exotic names and price with little more than the Black–Scholes–Merton formula.

Packages are combinations of vanillas (Chapter 12's spreads, straddles, and collars) sold as one contract and priced by adding up the pieces.

Non-standard American options, exercisable only in windows or with a strike that steps, need Chapter 20's lattice but no new theory.

A **gap option** pays $S_T - K_1$ if $S_T > K_2$ and nothing otherwise, which is a vanilla struck at K_2 plus $K_2 - K_1$ in a digital, so it is priced exactly and inherits the digital's hedging problem of Section 34.8; it appears where a payment is triggered at one level and measured from another, as in some insurance-linked and structured payoffs. A **shout option** lets the holder lock in the intrinsic value at one chosen date, and pays at expiration the greater of that locked value and the

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ordinary payoff: a cliquet (Section 34.12) whose single reset the holder chooses, sold mainly to investors who want to bank a gain without closing the position.

Compound options, options on options, have closed forms under Black–Scholes and are genuinely sensitive to volatility in a way vanillas are not: a compound option is a bet on volatility twice over, since both the outer and inner values depend on it, and the inner value at t_1 is set by whatever implied volatility prevails then, which today's surface does not fix. Their vega is correspondingly large relative to their price. Geske's formula for a call on a call (the right, on payment of K_1 at t_1, to acquire a call struck at K_2 expiring at T) is

$$c_{\text{call}} = S_0 e^{-qT} M(a_1, b_1; \rho) - K_2 e^{-rT} M(a_2, b_2; \rho) - K_1 e^{-rt_1} N(a_2), \qquad \rho = \sqrt{t_1/T} \quad (34.7)$$

where $M(\cdot, \cdot; \rho)$ is the bivariate standard normal distribution function,

$$a_1 = \frac{\ln(S_0/S^*) + (r - q + \sigma^2/2)t_1}{\sigma\sqrt{t_1}}, \qquad b_1 = \frac{\ln(S_0/K_2) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}},$$

$a_2 = a_1 - \sigma\sqrt{t_1}$, $b_2 = b_1 - \sigma\sqrt{T}$, and S^* is the asset price at t_1 at which the inner call is worth exactly K_1. That critical price is the only part of the formula requiring numerical work, and it is a one-dimensional root search on the Black–Scholes–Merton formula. The correlation entering M is $\sqrt{t_1/T}$ because one Brownian motion carries the price through both dates, which is the whole reason a compound option is a two-dimensional integral rather than a one-dimensional one. The other three compound options differ only in signs. Writing p_{call} for a put on a call, and c_{put} and p_{put} for a call and a put on a put,

$$\begin{aligned} p_{\text{call}} &= K_2 e^{-rT} M(-a_2, b_2; -\rho) - S_0 e^{-qT} M(-a_1, b_1; -\rho) + K_1 e^{-rt_1} N(-a_2) \\ c_{\text{put}} &= K_2 e^{-rT} M(-a_2, -b_2; \rho) - S_0 e^{-qT} M(-a_1, -b_1; \rho) - K_1 e^{-rt_1} N(-a_2) \\ p_{\text{put}} &= S_0 e^{-qT} M(a_1, -b_1; -\rho) - K_2 e^{-rT} M(a_2, -b_2; -\rho) + K_1 e^{-rt_1} N(a_2) \end{aligned}$$

Two things change with the inner option rather than the outer one. In the last two, S^* is the price at which the inner put is worth K_1, not the inner call, so it is a different root search; and a put loses value as the asset rises, so the outer option is exercised below S^* for a call on a put and above it for a put on a put.

Only two of the four are independent. Holding a call on a call and writing a put on the same call delivers that call at t_1 against K_1 paid, whatever the asset does in between, so

$$c_{\text{call}} - p_{\text{call}} = c(S_0, K_2, T) - K_1 e^{-rt_1}, \qquad c_{\text{put}} - p_{\text{put}} = p(S_0, K_2, T) - K_1 e^{-rt_1}$$

and those two identities are the cheapest check available on an implementation of any of the four.

Figure 34.1 puts the volatility claim in numbers. At a 10% volatility a call on a call is worth 3.02 against the underlying call's 6.81, and its value moves **2.2 times** as much for a given relative move in volatility; by 50% that ratio is down to 1.2. Both sides of the ratio are volatility elasticities, $\sigma\,(\partial V/\partial\sigma)/V$, the percentage change in value for a 1% change in volatility. The compound's elasticity barely moves: it sits near 1.1 throughout, so its value is close to proportional to volatility, while the vanilla's climbs from 0.50 to 0.86. Paying twice for volatility does not buy a larger vega; it buys a vega that stays large relative to a much smaller price. A compound option suits a firm bidding for a foreign-currency contract it may not win: it fixes the cost of the hedge if the bid succeeds and costs only the first premium if it fails.

Adding the second premium at its present value, $K_1 e^{-rt_1} = 4.88$, gives what ending up with the same call costs in total: 7.90 against the vanilla's 6.81 at a 10% volatility, 22.59 against 21.79 at 50%. The total exceeds the vanilla at every volatility (committing to exercise buys the identical instrument, so it cannot be cheaper), and the gap is what the right to walk away at t_1 is

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worth: **1.10** at 10%, narrowing to **0.80** at 50%, because the higher the volatility the less likely the holder is to decline.

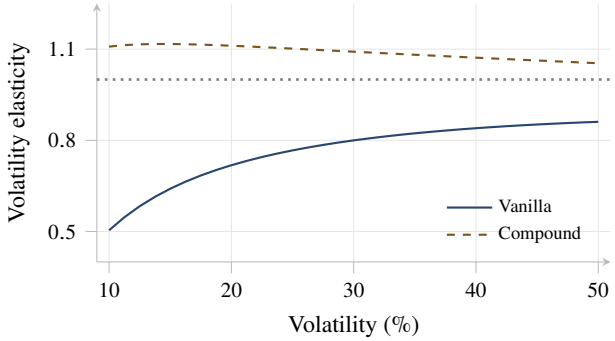


Figure 34.1: $\sigma\,(\partial V/\partial\sigma)/V$ for a call on a call and its underlying call, at $S_0 = K_2 = 100$, outer premium $K_1 = 5$ paid at $t_1 = 0.5$, $T = 1$, $r = 5\%$, no dividends.

Chooser options let the holder decide at t_1 whether the option is a call or a put, both struck at K and both expiring at T. A simple chooser, one whose call and put share a strike and an expiration as here, decomposes exactly into vanillas. Write $c(S,K,T)$ and $p(S,K,T)$ for Black–Scholes–Merton values. At the choice date the holder takes whichever is worth more, so the chooser is then worth

$$\max\bigl\{c(S_{t_1},K,T-t_1),\ p(S_{t_1},K,T-t_1)\bigr\}$$

Put–call parity at t_1 gives $p - c = Ke^{-r(T-t_1)} - S_{t_1}e^{-q(T-t_1)}$, the value of a short forward, which carries no optionality at all. Substituting it,

$$\begin{aligned}\max\{c,\ p\} &= c + \max\{0,\ p-c\}\\ &= c(S_{t_1},K,T-t_1) + e^{-q(T-t_1)}\max\Bigl(Ke^{-(r-q)(T-t_1)} - S_{t_1},\ 0\Bigr)\end{aligned}$$

where $e^{-q(T-t_1)}$ has been taken outside the maximum. Both terms are now instruments the vanilla market prices. The first is a call struck at K expiring at T, carried through the choice date untouched, and so worth $c(S_0,K,T)$ today. The second is $e^{-q(T-t_1)}$ puts struck at $Ke^{-(r-q)(T-t_1)}$, and they expire at t_1 rather than at T, because the maximum is taken on the price at the choice date. Hence

$$V_0 = c(S_0,K,T) + e^{-q(T-t_1)}\,p\Bigl(S_0,\ Ke^{-(r-q)(T-t_1)},\ t_1\Bigr) \tag{34.8}$$

Choosers are rare outright; their buyers are those who must commit to a volatility position before they know the direction.

Forward-start options are struck at the money on a future date. Under Black–Scholes their value scales with the spot, so an option struck at αS_{t_1} and expiring at T is worth

$$V_0 = e^{-qt_1}\,c(S_0,\ \alpha S_0,\ T-t_1) \tag{34.9}$$

At t_1 the option becomes a call struck at αS_{t_1} with $T-t_1$ to run. Scaling spot and strike together scales the Black–Scholes–Merton price in proportion, so the call is then worth $S_{t_1}\,c(1,\alpha,T-t_1)$, a number of units of the asset that is known today. A claim to one unit of the asset at t_1 is worth $S_0e^{-qt_1}$ now, the dividends before t_1 being forgone, and scaling back gives (34.9). The value is therefore a vanilla's price times e^{-qt_1}, with no dependence on anything observed between now and t_1. Under any realistic model the value is not this simple, because it depends on the *forward*

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smile (the smile quoted on a future date), and Section 21.3 measured how badly local volatility gets that wrong. A cliquet is a strip of forward-start options, so it is the purest instrument in this chapter for trading forward volatility, and the hardest to price for exactly the same reason.

34.3 BARRIER OPTIONS

Barriers are the largest exotic market, and they take up five sections of this chapter. A barrier option is a vanilla that is extinguished (**knock-out**) or created (**knock-in**) if the underlying touches a level H. With up and down, in and out, and calls and puts, there are eight basic types, and the in–out parity

$$\text{knock-in} + \text{knock-out} = \text{vanilla} \tag{34.10}$$

means only four need to be priced.

They exist because they are cheaper. A down-and-out call costs less than the vanilla because it can be extinguished, and a buyer who believes the barrier will not be touched is paying for the part of the distribution they think matters.

All eight have closed forms under Black–Scholes with continuous monitoring, and the whole family is assembled from four expressions. Write $\phi = +1$ for a call and -1 for a put, $\eta = +1$ for a down-barrier and -1 for an up-barrier, and set

$$\lambda = \frac{r - q + \sigma^2/2}{\sigma^2}, \qquad x_1 = \frac{\ln(S_0/K)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \qquad x_2 = \frac{\ln(S_0/H)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T},$$
$$y_1 = \frac{\ln\big(H^2/(S_0K)\big)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \qquad y_2 = \frac{\ln(H/S_0)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}.$$

Then

$$\begin{aligned}
A &= \phi S_0 e^{-qT} N(\phi x_1) - \phi K e^{-rT} N\big(\phi(x_1 - \sigma\sqrt{T})\big),\\
B &= \phi S_0 e^{-qT} N(\phi x_2) - \phi K e^{-rT} N\big(\phi(x_2 - \sigma\sqrt{T})\big),\\
C &= \phi S_0 e^{-qT} \left(\frac{H}{S_0}\right)^{2\lambda} N(\eta y_1) - \phi K e^{-rT} \left(\frac{H}{S_0}\right)^{2\lambda-2} N\big(\eta(y_1 - \sigma\sqrt{T})\big),\\
D &= \phi S_0 e^{-qT} \left(\frac{H}{S_0}\right)^{2\lambda} N(\eta y_2) - \phi K e^{-rT} \left(\frac{H}{S_0}\right)^{2\lambda-2} N\big(\eta(y_2 - \sigma\sqrt{T})\big),
\end{aligned} \tag{34.11}$$

and the four knock-in values are read off the table below. Knock-outs follow from the parity (34.10).

Knock-in option	Value, by where the barrier sits relative to the strike			
Down-and-in call	$H \le K$:	C	$H > K$:	$A - B + D$
Up-and-in put	$H \ge K$:	C	$H < K$:	$A - B + D$
Up-and-in call	$H \ge K$:	$B - C + D$	$H < K$:	A
Down-and-in put	$H \le K$:	$B - C + D$	$H > K$:	A

Three things can be read off (34.11) directly. A is the vanilla, since $x_1 = d_1$; the barrier enters only through B, C, and D. B is A with the barrier replacing the strike inside the normal arguments: the strike still sets the payoff, but the barrier now sets the boundary. C and D are the reflections of A and B in the barrier, weighted by the powers of H/S_0 that the reflection principle produces. That principle pairs each path that touches the barrier with its mirror image in the barrier from the moment of touching. When the price has a drift the mirror image is not equally likely; the powers correct for that, and it is why Section 34.5's static hedge is exact only when the drift is zero.

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The two entries equal to A are the vanilla itself, because a barrier lying between the spot and the strike is one the option cannot finish in the money without having crossed: an up-and-in call with $H < K$ knocks in on every path that pays it anything.

The formula also contains the static hedge. When $r = q$ the exponent λ collapses to $1/2$, the two weights become H/S_0 and S_0/H, and the down-and-in call's C reads term for term as K/H puts struck at H^2/K. Section 34.5's reflection result is visible in the formula, which is a good reason to write the formula out rather than call a library.

The formulas assume no **rebate**, a fixed amount R paid to the holder when a knock-out option is extinguished. A rebate is valued separately, and for a down barrier both versions have closed forms in the same λ as (34.11). If it is paid at expiration it is a digital on the running minimum, paying if $m_T \le H$, and is worth Re^{-rT} times the risk-neutral probability that the barrier is touched,

$$P(m_T \le H) = N\left(\frac{\ln(H/S_0)}{\sigma\sqrt{T}} - (\lambda-1)\sigma\sqrt{T}\right) + \left(\frac{H}{S_0}\right)^{2\lambda-2} N\left(\frac{\ln(H/S_0)}{\sigma\sqrt{T}} + (\lambda-1)\sigma\sqrt{T}\right)$$

whose weight $(H/S_0)^{2\lambda-2}$ is the one that already appears in C and D. If it is paid on the touch it is a first-passage claim, discounted from a random date, and is worth

$$R\left[\left(\frac{H}{S_0}\right)^{\lambda-1+\psi} N(z) + \left(\frac{H}{S_0}\right)^{\lambda-1-\psi} N\big(z - 2\psi\sigma\sqrt{T}\big)\right]$$

where

$$\psi = \sqrt{(\lambda-1)^2 + \frac{2r}{\sigma^2}}, \qquad z = \frac{\ln(H/S_0)}{\sigma\sqrt{T}} + \psi\sigma\sqrt{T}$$

An up barrier reverses the signs of the arguments of N in both. For the down-and-out call of Example 34.1 below, $\lambda = 1.3$, and with no dividends $\psi = \lambda$ exactly; the barrier is touched with probability 0.652, so a rebate of 1 paid at expiration is worth $e^{-0.05} \times 0.652 = 0.620$, and the same rebate paid on the touch, which arrives earlier on average, is worth 0.643. Either is added to the no-rebate value to price the contract with it.

Example 34.1 What the barrier removes

$S_0 = 100$, $K = 100$, $H = 90$, $r = 5\%$, $\sigma = 25\%$, $T = 1$, continuously monitored.

Vanilla call	12.335999
Down-and-out call	9.111221
Down-and-in call (the difference)	3.224778

The down-and-in value is the first row of the knock-in table, the single expression C with $\lambda = 1.3$: $(H/S_0)^{2\lambda} = 0.760380$ and $(H/S_0)^{2\lambda-2} = 0.938740$, $y_1 = -0.517884$, and

$$C = 100 \times 0.760380 N(-0.517884) - 100e^{-0.05} \times 0.938740 N(-0.767884) = 3.224778$$

The barrier removes 26.1% of the premium—a discount large enough to be the entire reason the instrument is bought. Two observations shape everything after this.

The knock-out is not simply a cheaper call. Approaching the barrier makes the option *less* valuable while approaching the strike makes it more valuable, and between those two pulls a knock-out's delta can exceed one in magnitude and its gamma can change sign. Figure 34.4, in Section 34.8, measures both. A **reverse knock-out** (barrier above the strike, so the option is knocked out when it is deep in the money) has a discontinuous payoff at the barrier and is among the most difficult vanilla-like instruments in the market.

The price depends on the whole path distribution. The vanilla depends only on the terminal distribution, which Section 19.2 showed the option chain reveals. The barrier depends on the probability of touching, which the chain does not reveal.

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34.4 WHY BARRIERS BREAK TREES

Section 20.4 promised this, and the failure is instructive rather than merely inconvenient.

Example 34.2 More steps, worse answer

Price Example 34.1's down-and-out call ($S_0 = K = 100$, $H = 90$, $r = 5\%$, $\sigma = 25\%$, $T = 1$, no dividends, European exercise) on a Cox–Ross–Rubinstein (CRR) tree, knocking out at any node at or below H. The tree can knock out only at a time step, while the analytic answer is the continuously monitored one: **9.111221**.

Steps	Tree	Error	Effective barrier
50	9.142046	+0.031	89.94
51	10.494157	**+1.383**	86.93
100	9.944477	+0.833	88.25
200	9.144117	+0.033	89.94
400	9.438414	+0.327	89.36
1600	9.153506	+0.042	89.92

Fifty-one steps is forty-five times worse than fifty. And 1600 steps is worse than 200.

The mechanism is entirely in the last column. The tree can only knock out at a node, so the barrier it actually enforces is the highest node level at or below H. As n changes, that level jumps around: at $n = 50$ it is 89.94 and at $n = 51$ it is 86.93. The tree is pricing a different contract at every step count, and the error is dominated by which contract rather than by discretization.

The error is also one-signed. The effective barrier is always at or below the true one, so the option is always overpriced. That is worse than a random error: averaging two grids does not help (the binomial error of a vanilla alternates in sign with the parity of n, which is why averaging neighboring grids works there; here it never changes sign) and Richardson extrapolation, which Section 20.3 already warned about, makes it worse.

Now fix it. Choose n so that a layer of nodes falls on the barrier, to within the rounding of n. In a CRR tree the node levels are $S_0 u^{\pm j}$ with $u = e^{\sigma\sqrt{T/n}}$, so this requires

$$\frac{\ln(S_0/H)}{\sigma\sqrt{T/n}} = m \quad \text{for an integer } m \qquad \Longrightarrow \qquad n = \left\lfloor \frac{m^2\sigma^2 T}{\ln^2(S_0/H)} \right\rfloor$$

m	Steps	Tree	Error
4	90	9.111965	+0.00074
6	202	9.119059	+0.00784
8	360	9.113142	+0.00192
9	456	9.111153	−0.00007
12	810	9.113370	+0.00215
16	1441	9.112680	+0.00146

At 90 steps with the layer on the barrier, the error is 0.0007. At 100 steps without it, the error is 0.833—more than a thousand times larger, with more steps.

The lesson generalizes well beyond barriers. A numerical method that resolves the payoff's discontinuities converges; one that does not, does not—and refining a grid that misses the discontinuity refines the wrong problem.

Never let a user choose the step count on a barrier product. The pricing library should choose it from the barrier.

In Figure 34.2's left panel, the sawtooth is not noise: each descending run is the effective barrier drifting up toward H as n increases, and each jump is a node crossing the barrier and resetting it.

The alternative is a method whose grid is free. The binomial tree's trouble is that it has no

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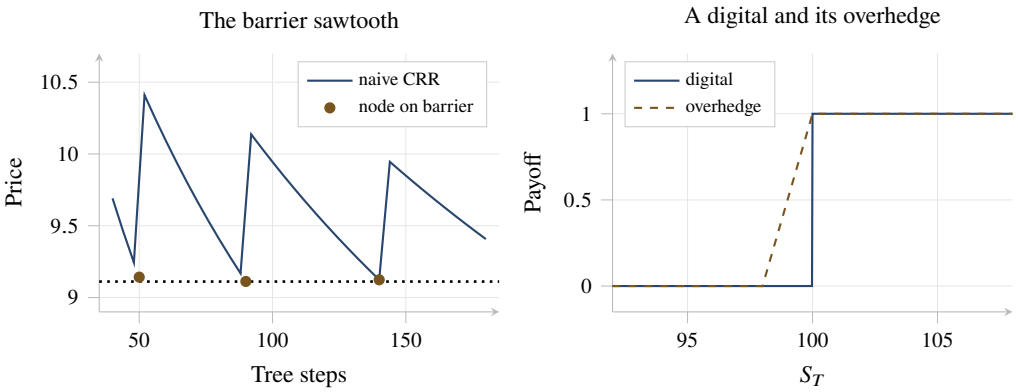


Figure 34.2: Left: Example 34.2; the dotted line is the analytic value. Right: a one-sided call spread dominating a digital.

spare parameter: $u = e^{\sigma\sqrt{\Delta t}}$ is fixed once σ and Δt are, so the node levels are determined and the barrier falls where it falls. A trinomial tree carries a space step that can be set independently of the time step, so choosing $\Delta x = \ln(S_0/H)/m$ for an integer m puts a layer of nodes on the barrier at *every* step count rather than at lucky ones. In an implicit finite-difference scheme the barrier is made the lower boundary of the grid, where the knock-out is the exact condition $V = 0$ rather than a test applied at interior nodes, and the space step is refined as far as wanted independently of the time step. That is the sense in which these are the natural methods for barriers: the discontinuity is put on the mesh by construction instead of being hunted for. On Example 34.2's contract the trinomial tree's largest error over 50, 51, 100, 200 and 400 steps is **0.017**, against the binomial's **1.383**, and the jump from 50 steps to 51 that cost 1.35 becomes 0.02. This is why barrier books are priced on grids rather than on binomial trees.

34.5 STATIC REPLICATION

Section 18.13 argued that where gamma diverges, a static hedge is the only workable answer. Barriers are the main place that argument pays off.

Example 34.3 A barrier hedged with vanillas and never touched again

When the drift is zero ($r = q$), a **reflection** argument gives an exact static replication. A down-and-in call with strike K and barrier $H < K$ has the same value as K/H puts struck at H^2/K.

The argument has two cases. If the barrier is never touched, the call never knocks in, and the puts expire worthless too: they are struck at H^2/K, below H, and the price has stayed above H. If the barrier is touched, then at that moment K/H puts struck at H^2/K are worth exactly one call struck at K (a symmetry of the lognormal distribution that holds when $r = q$), so the puts are sold and the call bought at no cost, and the hedge has become the option.

Nothing has to be computed to confirm it, because (34.11) contains it. A down-and-in call with $H \le K$ is the single block C, and at $r = q$ the exponent is $\lambda = 1/2$, so the two weights collapse to H/S_0 and S_0/H:

$$C = e^{-rT}\left[H\,N(y_1) - K\frac{S_0}{H}\,N\big(y_1 - \sigma\sqrt{T}\big)\right], \qquad y_1 = \frac{\ln\big(H^2/(S_0K)\big)}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2}$$

Now price the puts. One struck at H^2/K has $d_1 = \ln\big(S_0K/H^2\big)/(\sigma\sqrt{T}) + \sigma\sqrt{T}/2$, so $-d_1 = y_1 - \sigma\sqrt{T}$ and $-d_2 = y_1$, and K/H of them are worth

$$\frac{K}{H}\left[\frac{H^2}{K}e^{-rT}N(y_1) - S_0e^{-rT}N\big(y_1 - \sigma\sqrt{T}\big)\right] = e^{-rT}\left[H\,N(y_1) - K\frac{S_0}{H}\,N\big(y_1 - \sigma\sqrt{T}\big)\right]$$

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term for term the same expression. The replication is therefore exact rather than close, at any common level of $r = q$: the bracket contains no r, so raising the common rate only discounts the same number. The condition is zero *drift*, not zero rates.

Now break the condition. At $r = 5\%$, $q = 0$:

Down-and-in call	3.224778
Replicating puts	1.861532
Static hedge error	**−42.3%**

The replication is not robust to drift, and 42% is not a rounding error. Refinements exist (replicating with a strip of vanillas at several maturities rather than one, which trades exactness for robustness), and they are what desks actually use.

But the reason to want a static hedge survives the error. A static hedge is a portfolio of vanillas set up once and traded again at most once, when the barrier is touched. It requires *no rebalancing*, so it has no running transaction costs and no gamma at the barrier; and it does not depend on the surface dynamics that Section 19.12 said barrier prices depend on. A hedge that is wrong by a known amount and never has to be adjusted can be better than one that is right in a model and must be adjusted precisely where the model is least reliable.

34.6 DISCRETE MONITORING

Every formula so far assumes the barrier is monitored continuously. *Few equity contracts are*: their barriers are observed at a daily close, sometimes weekly or monthly; only FX barriers, watched around the clock, come close. The gap is larger than it looks and has a clean correction.

Example 34.4 What a daily close is worth

Broadie, Glasserman, and Kou showed that a barrier monitored m times behaves, to first order, like a *continuous* barrier shifted *away* from the spot, and pricing with the shifted barrier is the **continuity correction**:

$$H_{\text{eff}} = H\, e^{\pm\beta\sigma\sqrt{T/m}}, \qquad \beta = -\frac{\zeta(1/2)}{\sqrt{2\pi}} = 0.5826 \tag{34.12}$$

where ζ is the Riemann zeta function, $\zeta(1/2) = -1.4604$, and the sign in the exponent is chosen so the barrier moves further from the current spot: down for a down-barrier, up for an up-barrier.

Applied to Example 34.1's down-and-out call, whose continuous value is 9.111221:

Monitoring	m	**Effective** H	**Price**	**vs. continuous**
Daily	252	89.18	9.527218	+4.57%
Weekly	52	88.20	9.968231	+9.41%
Monthly	12	86.29	10.675121	+17.16%
Quarterly	4	83.68	11.365166	+24.74%

Even daily monitoring is worth 4.6% of the option, and monthly monitoring 17%. A knock-out priced as continuous is undervalued, because a path that dips below the barrier between closes is not caught.

Verify it independently. Simulate the daily-monitored contract directly with 200,000 antithetic pairs:

	Value	**Against Monte Carlo**
Monte Carlo (daily)	9.523044	(s.e. 0.0243)
Corrected formula (34.12)	9.527218	+0.17 standard errors
Uncorrected (continuous)	9.111221	**−16.9 standard errors**

The correction lands within a fifth of a standard error and the uncorrected formula is nearly seventeen away. The correction looks ad hoc and is not: β comes from the expected overshoot of a

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random walk past a level, and the same constant appears wherever a discretely observed process is compared with a continuous one. Three consequences for practice follow.

The correction is not optional at low monitoring frequencies. Quarterly observation moves the price by a quarter of its value, which no bid–ask spread absorbs.

It interacts with Section 34.4. A tree with a node layer on the *stated* barrier prices the continuous contract; to price the discrete one, the layer goes on H_{eff}. A misaligned layer and the correction both move the barrier away from spot, so an unaligned tree that skips the correction can land near the discrete price by accident: on this contract 76 steps give 9.532 against 9.527, while 100 steps are 0.42 high and 50 are 0.39 low. Neither adjustment substitutes for the other, and a tree that is right for the wrong reason at one step count is wrong at the next.

It matters most for the products of Chapter 35. An autocallable (a note that redeems early if the underlying is above a call barrier on an observation date, and exposes the investor to the underlying's loss if a lower knock-in barrier is breached) observes its knock-in barrier daily and its call barrier monthly, so a single instrument carries two different monitoring frequencies and needs two different corrections.

34.7 BARRIERS AND THE VOLATILITY SURFACE

A barrier price depends on the volatility surface in a way a vanilla price does not, and Section 21.11 measured by how much. Two models, local volatility and a stochastic-volatility model, were calibrated to the *same* vanilla surface and verified to reproduce it. They then priced a one-year up-and-out call, spot and strike at 100, barrier at 110, monitored daily: 0.8402 under the stochastic-volatility model against 0.4244 under local volatility, a difference of **49.5%** on an option worth less than a point.

The design of that experiment is Hull and Suo's (2002). What the spread measures is not an error in either model—both are right about every quoted vanilla—but the size of the question the market has left unanswered.

The reason for the spread is Section 19.12's. A vanilla depends on the terminal distribution and a barrier depends on the path, and two models can agree on every terminal distribution while disagreeing about how the underlying gets there. Calibration to vanillas is therefore necessary and not sufficient, and a desk that reports a barrier price to four decimals from a calibrated model is reporting the model. Three practical responses follow, in ascending order of what they give up.

Price with the model, reserve against the range. Compute under two or three models and hold the spread as a reserve. Section 21.11's argument for reporting model risk applies exactly.

Hedge statically where possible, which converts the model question into a residual and a known error, as Section 34.5 described.

Or refuse the trade. A reverse knock-out at a barrier where the two models differ by half the premium is a trade whose price the desk does not know, and there are market conditions in which the correct response is to say so.

This is also where Section 18.6's vanna and volga enter. A knock-out's vega can change sign across the spot range, so its value depends on how vega moves with spot and with volatility, which a single flat volatility cannot price. Foreign-exchange desks therefore quote barriers by the vanna–volga rule: the Black–Scholes–Merton value plus the cost of the option's vanna and volga, each priced at what the risk reversal and the butterfly charge for that exposure. It is a quoting convention rather than a model, and it is why a barrier quote moves when the wings of the smile move.

34.8 DIGITALS AND THE OVERHEDGE

This section treats digitals at more length than their own market would justify, because a digital is the simplest payoff with a jump in it. Everything that makes a discontinuous payoff hard

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to hedge dynamically appears here with nothing else in the way: a delta that grows without bound as expiration approaches, a hedge that has to be overhedged rather than replicated, and a bid–ask spread set by what the overhedge costs. A barrier option carries the same jump at its barrier, which is why Example 34.6 returns to the argument for the reverse knock-out.

Section 12.8 showed a digital has no finite replicating portfolio. Here is what that costs.

The closed forms are the two halves of the Black–Scholes–Merton formula taken apart. A **cash-or-nothing** call pays one unit of cash if $S_T > K$; an **asset-or-nothing** call pays S_T in the same event. Their values are

$$V_{\text{cash}} = e^{-rT}N(d_2), \qquad V_{\text{asset}} = S_0 e^{-qT}N(d_1) \tag{34.13}$$

with d_1 and d_2 the usual Black–Scholes arguments, and the put versions replacing $N(\cdot)$ with $N(-\cdot)$. The vanilla call is $V_{\text{asset}} - K\,V_{\text{cash}}$, which is Section 15.4's reading of $N(d_1)$ and $N(d_2)$ restated as two instruments rather than two integrals.

The formula is also the limit of the call spreads the example below prices, writing $c(K)$ for the value of a call struck at K:

$$\frac{c(K-\varepsilon/2) - c(K+\varepsilon/2)}{\varepsilon} \xrightarrow[\varepsilon\to 0]{} -\frac{\partial c}{\partial K} = e^{-rT}N(d_2) \tag{34.14}$$

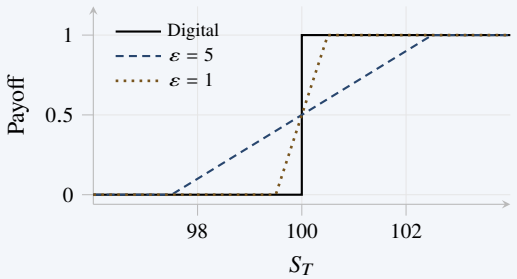
A digital is the strike-derivative of the vanilla, one differentiation short of Section 19.2's implied density, which is the second. The price is therefore a derivative of quoted prices, and a derivative of a quoted price is not itself quoted.

Example 34.5 What certainty costs

A cash-or-nothing call paying \$1 if $S_T > 100$, with $S_0 = 100$, $r = 5\%$, $\sigma = 25\%$, $T = 1$ and no dividends: worth **0.504049**.

Replicate with a call spread of width ε straddling the strike (long $K - \varepsilon/2$, short $K + \varepsilon/2$), held in quantity $1/\varepsilon$. Each spread pays at most ε, so $1/\varepsilon$ of them pay nothing below $K - \varepsilon/2$, the digital's full dollar above $K + \varepsilon/2$, and rise linearly between them. That is (34.14) evaluated at finite ε:

ε	Spreads	Value
5.00	0.2	0.504159
1.00	1.0	0.504054
0.10	10.0	0.504050
0.01	100.0	0.504049



The value converges and the position diverges. The plot shows why the value converges (the payoff approaches the step) and also why this spread is not yet a hedge: it crosses the digital, paying more below the strike and less above it.

What the dealer has sold has no finite Greeks. With τ the time remaining to expiration, differentiating $e^{-r\tau}N(d_2)$ once and twice, and collecting $d_1 = d_2 + \sigma\sqrt{\tau}$,

$$\Delta = \frac{e^{-r\tau}N'(d_2)}{S\sigma\sqrt{\tau}}, \qquad \Gamma = -\frac{e^{-r\tau}N'(d_2)\,d_1}{S^2\sigma^2\tau}$$

At the money $d_1 = 1.3\,\sigma\sqrt{\tau}$, since $(r - q + \sigma^2/2)/\sigma^2 = 1.3$ here, so both Greeks grow like $1/\sqrt{\tau}$: each doubles whenever the time remaining is quartered. Away from the money it is the d_1 in the numerator that carries the gamma's sign, so the gamma changes sign at the delta's peak, $Ke^{-(r-q+\sigma^2/2)\tau}$. With time counted in trading days (252 a year, of six and a half hours), that point closes in on the strike as expiration approaches (three cents below it with a day to run, under a

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hundredth of a cent with a minute), and with a minute to run $|\Gamma|$ five cents to either side of it reaches 32. The delta is 0.25 with a day to run, 0.65 with an hour, 5.0 with a minute and 38.8 with a second.

Holding $1/\varepsilon$ spreads caps the delta at $1/\varepsilon$, which is the whole reason for holding spreads rather than the digital; and even at that cap, with $\varepsilon = 0.10$, ten shares of stock have to be traded for every dollar of payoff while the price sits inside the window in the last moments before expiration, with no time to trade them. That is what Section 18.13's "gamma diverges" means when someone has to do it.

So dealers overhedge. To **overhedge** is to hedge with a position whose payoff is at least the sold option's everywhere, paying a known premium in place of an unhedgeable jump. Sell the digital and buy $1/\varepsilon$ call spreads struck at $K - \varepsilon$ and K, so that the upper strike stays at the digital's strike and the whole window lies *below* it. That position already pays the full dollar at K and something below it, so it dominates the digital's payoff everywhere. The table widens the window and compares each cost with the digital's 0.504049:

ε	Strikes	Cost	Premium over the digital
1	[99, 100]	0.511635	+1.5%
2	[98, 100]	0.519249	+3.0%
5	[95, 100]	0.542210	+7.6%

That premium is what certainty costs. A dealer who will not carry the residual cannot sell the digital below the dominating spread, nor buy it above the mirror image: a spread struck entirely *above* the strike, which the digital dominates. At a one-point window those bounds are 0.4965 and 0.5116 around a digital worth 0.5040, so no two-way price narrower than **3.0%** is hedged on both sides. What a desk actually quotes is wider: under a skew the digital's own value moves (Example 19.3 differentiates along the smile), every one of the $1/\varepsilon$ spreads is bought at the ask, and the overhedged book ties up capital. None of these is in the model here; this is the floor the payoff alone puts under the quote. Figure 34.2's right panel shows the dominance.

The width is a business decision, not a modeling one. A tighter spread is a better price, a worse hedge, and a higher cap on the position the desk may have to trade: one share at a point wide, ten at a tenth, a hundred at a hundredth. The dealer chooses where on that trade-off to sit based on how much of the position they expect to hold to expiration.

This generalizes to every discontinuous payoff in the chapter that has a vanilla market beneath it. A reverse knock-out is a digital in disguise. It pays a positive amount right up to the barrier and nothing beyond it, the discontinuity Section 34.3 pointed to. The overhedge for it is a barrier struck slightly further away, and the premium is much larger, as Example 34.6 shows.

Who trades digitals. Very little of the market trades a digital on its own. Most digital risk sits inside other products as a coupon trigger: each observation date on an autocallable pays its coupon only if the underlying is above a level, and a range accrual pays a day's interest only if a rate stays inside a band, so both are strips of digitals, held by the investors of Chapters 33 and 35 and sold by the issuing dealers; held above the trigger, a digital loses value as volatility rises, which is how those investors come to be short volatility. The one active standalone market is in foreign exchange, where European digitals and their path-dependent relatives (the **one-touch**, which pays if the rate trades at a level at any time before expiration, and the **no-touch**, which pays if it never does) are standard dealer products. Macro hedge funds buy them to take a view on a level for a fraction of a vanilla's premium.

Professionals have little need for a listed digital, because a tight call spread already is one. Cboe listed binary options, as digitals are also called, on the S&P 500 and the VIX in 2008, and they drew almost no volume. Binaries found their buyers among retail investors instead, sold by offshore platforms that were the counterparty to every trade, usually at maturities of minutes or hours. The European Securities and Markets Authority restricted their sale to retail clients from July 2018 (a measure renewed in three-month steps and later carried into national

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rules), and Israel closed its binary-options industry outright.

The growth since 2024 has come from **event contracts**: cash-or-nothing contracts on an inflation release, a central-bank decision, an election or, for most of the volume since 2025, a sports result, listed on exchanges regulated by the Commodity Futures Trading Commission and offered on prediction markets; as of this writing the venues were still in dispute with the states that regard sports contracts as gambling. They are digitals in the strict sense of this section, but on outcomes that no vanilla option spans. There is no call spread to overhedge them with, so Example 34.5's argument does not reach them: the price is a probability set by the order book, and the market maker carries the jump at the event rather than hedging it.

Example 34.6 The instrument that is short its own underlying

An up-and-out call struck at 100 with the barrier at 120, one year, 25% volatility, and $r = q = 0$: zero carry, so the reflection results of Section 34.5 apply exactly. Its value is the vanilla A less the up-and-in call $B - C + D$ from the third row of Section 34.3's table, by in–out parity. The vanilla is worth 9.9476; the knock-out is worth **0.6709**, *6.7% of it*. Almost the whole premium has been given away, because the option is extinguished in precisely the states where a call is valuable.

Its delta is the striking part:

Spot	Value	Delta
100	0.6709	−0.0149
110	0.4081	−0.0359
115	0.2132	−0.0414
118	0.0863	−0.0430

The delta is negative from a spot of 94.3 upward. This is a call option whose holder loses money when the underlying rises, because approaching 120 removes more value through the knock-out than it adds through moneyness. Its holder hedges by buying stock, as though short a call—the reverse of every intuition built up in Chapter 18—and the sign flips back the moment the barrier is removed.

The overhedge is where the difficulty becomes a price. Dominating the payoff means pricing and hedging the contract as a knock-out with the barrier further away, and the cost rises steeply:

Barrier used	Value	Cost over the contract
122	0.8947	+33.4%
124	1.1513	+71.6%
126	1.4377	+114.3%

Moving the barrier out by two points, 1.7% of its level, costs a third of the option's value, against 7.6% for the digital's much wider five-point window in Example 34.5. That is why a reverse knock-out's bid–ask spread runs to percentage points while a down-and-out's runs to basis points, and it is a property of the payoff rather than of any model: the cliff is 20 points high on an option worth 0.67.

Figure 34.3 plots this option. The value is a hump whose peak sits at 94.3, six points *below* the strike, because above that level a rise in spot removes more value by raising the probability of a knock-out than it adds by moving the call further into the money, and the delta in the right panel is simply the slope of the left one, crossing zero exactly at the peak. A call option with an interior maximum is not a call option in any sense Chapter 18 would recognize, and the hedging difficulty is that the maximum is inside the trading range rather than at the edge of it.

Figure 34.4 takes the same option with a few weeks to run rather than a year, and adds the gamma. The delta climbs to 0.67 by a spot of 104, turns down, crosses −1 at 115.7, and reaches −1.46 at the barrier, where a vanilla's delta is +1; with a year to run it never left $[-0.043,\ 0.027]$. The gamma is negative above 104.4, where a vanilla's is positive everywhere,

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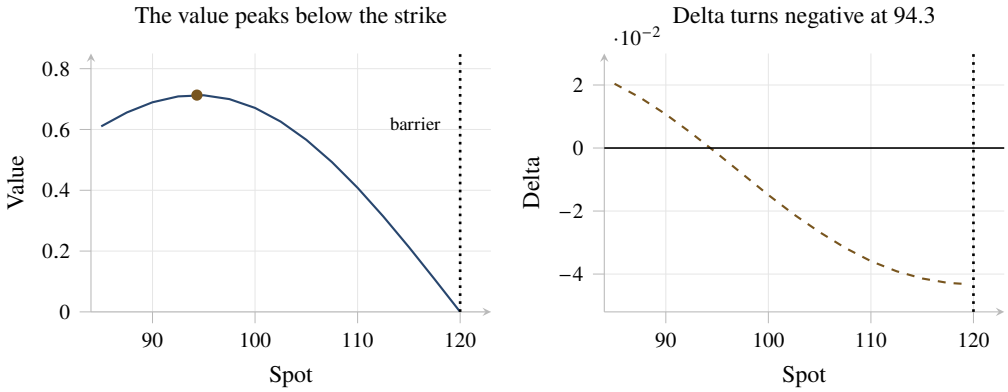


Figure 34.3: Example 34.6's up-and-out call: $K = 100$, $H = 120$, $\sigma = 25\%$, $T = 1$, $r = q = 0$.

and at -0.23 near a spot of 113 its magnitude is three times the largest gamma the vanilla ever has.

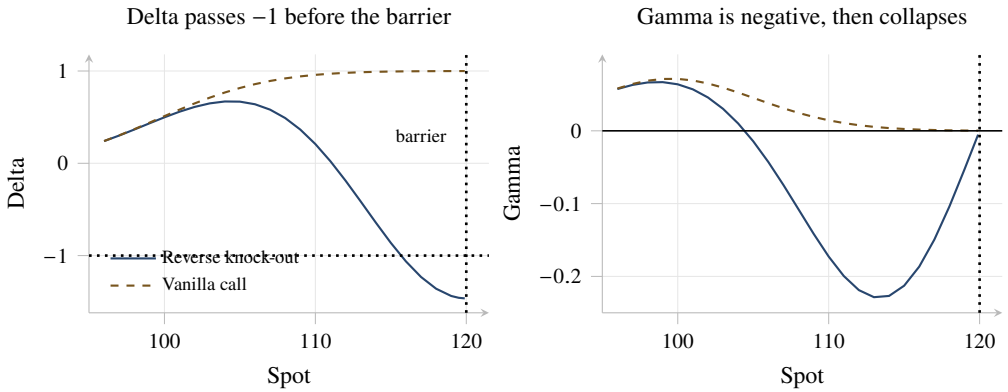


Figure 34.4: Up-and-out call against a vanilla call: $K = 100$, $H = 120$, $\sigma = 25\%$, $T = 0.05$, $r = q = 0$. The dotted vertical line is the barrier, the dotted horizontal one $\Delta = -1$.

The hedging problem is in the last few points of the right-hand panel. Approaching the barrier the value becomes nearly linear in the spot (0.146 at a spot of 119.9, against 0.146 on the line of slope -1.46 through the barrier), so the gamma collapses to -0.005 while the delta does not: it is still -1.46. A hedge of 1.46 shares against an option worth 0.15 means a tenth of a point of spot moves the hedge by the option's entire value, and at the barrier the option ceases to exist, so that delta has to be unwound at once. Gamma, which normally warns a hedger that rebalancing is about to become expensive, is closest to zero exactly where the position is most dangerous. In the terms of Chapter 18's expansion, equation (18.1), the gamma term is what announces that the delta term's coefficient is about to change, and here it is silent because the change is not gradual: at the barrier the delta jumps from -1.46 to zero, which no second-order term of an expansion around a smooth value can represent.

34.9 LOOKBACKS AND ASIANS

This section compares two path-dependent payoffs, one cheaper than the vanilla and one dearer. Both have closed forms, the Asian only in its geometric version, and those come first.

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Two averages are in use. The traded contract takes the arithmetic average of its fixings, $\frac{1}{n}\sum_{i=1}^{n} S_{t_i}$, the discrete form of the A_T in (34.3). The **geometric average** multiplies the fixings and takes the nth root,

$$G_T = \Big(\prod_{i=1}^{n} S_{t_i}\Big)^{1/n} = \exp\Big(\frac{1}{n}\sum_{i=1}^{n} \ln S_{t_i}\Big),$$

which becomes $G_T = \exp\left(\frac{1}{T}\int_0^T \ln S_t\, dt\right)$ when the averaging is continuous, and which never exceeds the arithmetic average of the same prices. Its logarithm is an average of normal variables and is therefore normal, so the geometric-average Asian call, paying $(G_T - K)^+$, is exactly lognormal: it is the Black–Scholes–Merton formula with two substitutions (Kemna and Vorst). Averaging continuously over $[0, T]$,

$$c_G = S_0 e^{(b_G - r)T} N(d_1) - K e^{-rT} N(d_2), \qquad \sigma_G = \frac{\sigma}{\sqrt{3}}, \qquad b_G = \frac{1}{2}\left(r - q - \frac{\sigma^2}{6}\right) \tag{34.15}$$

with $d_1 = \left[\ln(S_0/K) + (b_G + \sigma_G^2/2)T\right]/(\sigma_G\sqrt{T})$ and $d_2 = d_1 - \sigma_G\sqrt{T}$. The $1/\sqrt{3}$ is Section 33.8's volatility multiplier, the square root of its variance ratio; b_G is the drift the geometric average inherits: half of $r - q$, because an average over $[0, T]$ sits halfway along the path, less $\sigma^2/12$, the amount by which a geometric mean sits below an arithmetic one.

The arithmetic-average Asian, the one actually traded, has no closed form, because a sum of lognormals is not lognormal. Two devices cover the gap. Matching the average's first two moments to a lognormal and pricing with Black's formula gives 6.883 for the contract in Example 34.7, against 6.85 by simulation: about half a percent high, which is the size of error the approximation literature debates, though it grows with volatility, to 1.4% at 50%. The arithmetic version is worth 5% more than the geometric one priced in that example, the gap between an arithmetic and a geometric mean, and that gap rises roughly in proportion to volatility, from 2% at a 10% volatility to 11% at 50%. The second device is (34.15), which serves as a control variate for simulation: the two payoffs correlate at 0.9994, and subtracting the geometric payoff one for one cuts the variance by a factor of 330, or about 850 with the regression-optimal coefficient. The instrument that is traded is the one without a formula, and it barely matters—which is the reverse of the situation for every barrier in this chapter.

The floating-strike lookback call paying $S_T - S_{\min}$, where $S_{\min}$ is the running minimum m_t observed to date (S_0 for a new contract), is worth (Goldman, Sosin and Gatto)

$$c_{\text{lb}} = S_0 e^{-qT} N(a_1) - S_0 e^{-qT}\,\kappa\, N(-a_1) - S_{\min} e^{-rT}\left[N(a_2) - \kappa\, e^{Y_1} N(-a_3)\right], \qquad \kappa = \frac{\sigma^2}{2(r-q)} \tag{34.16}$$

where

$$a_1 = \frac{\ln(S_0/S_{\min}) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad a_3 = \frac{\ln(S_0/S_{\min}) + (-r + q + \sigma^2/2)T}{\sigma\sqrt{T}},$$

$a_2 = a_1 - \sigma\sqrt{T}$, and $Y_1 = -2(r - q - \sigma^2/2)\ln(S_0/S_{\min})/\sigma^2$. The floating-strike put, paying $S_{\max} - S_T$ with $S_{\max}$ the running maximum, is the mirror image: the maximum takes the place of the minimum and the drift terms change sign. The formula is singular at $r = q$ (κ diverges), though the value is not, the limit being finite; an implementation that does not special-case zero drift will return nonsense on precisely the parameter set Section 34.5 cares about.

Example 34.7 Averaging destroys optionality; looking back manufactures it

All at $S_0 = K = 100$, $\sigma = 25\%$, $r = 5\%$, $T = 1$:

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	Value	Relative to vanilla
Vanilla at-the-money call	12.3360	1.00
Floating-strike lookback call	20.5522	**1.67**
Geometric-average Asian call	6.5298	**0.53**

The underlying and the volatility are the same, and the two exotics differ by a factor of three.

The Asian is cheap because an average has less variance than an endpoint. Section 33.8 computed the ratio: $(n+1)(2n+1)/(6n^2)$, tending to $1/3$, so the effective volatility is about $1/\sqrt{3}$ of the true one. An Asian option is a vanilla on a less volatile quantity, and that explains most of the discount: at $25\%/\sqrt{3} = 14.4\%$ an at-the-money call is worth 8.38, about two-thirds of the vanilla. The rest comes from the average's lower forward, the drift b_G of (34.15), which takes the ratio to 0.53.

The lookback is expensive because it removes the timing decision. A floating-strike lookback call pays $S_T - m_T$, so the holder buys at the best price achieved. With nothing left to get wrong, the lookback is worth 67% more than the vanilla.

Figure 34.5 repeats the comparison across volatilities, and it separates what is optionality from what is not. At zero volatility the vanilla and the lookback are worth the same 4.88: the path then rises steadily at the 5% rate, its minimum is the starting price, and both pay only the drift. The Asian starts lower, at 2.41, because the average of a rising path sits below its end, and that gap is carry rather than optionality. Between 25% and 50%, each volatility point adds about 0.61 to the lookback, 0.38 to the vanilla and 0.18 to the Asian. Three practical notes follow.

Asians dominate commercial usage and lookbacks barely exist. A company hedging a year of purchases has an average exposure and wants an Asian; no one has an economic exposure to the minimum of a price path. Lookbacks are taught more than they are traded.

The Asian's model risk is low and the lookback's is high. An average smooths the path, so an Asian is less sensitive to the path distribution than the vanilla is to the terminal one; a lookback is a bet on the extremes and is very sensitive to it.

Discrete monitoring matters in both, in opposite directions. A monthly-averaged Asian is worth more than a continuously averaged one (fewer observations, more variance), while a daily-monitored lookback is worth less than a continuous one, because a discretely observed minimum is higher. For the lookback, Section 34.6's continuity correction is the same adjustment, its sign following from which way discretization moves the observed extreme; for the Asian it is the variance ratio above, evaluated at the actual number of fixings.

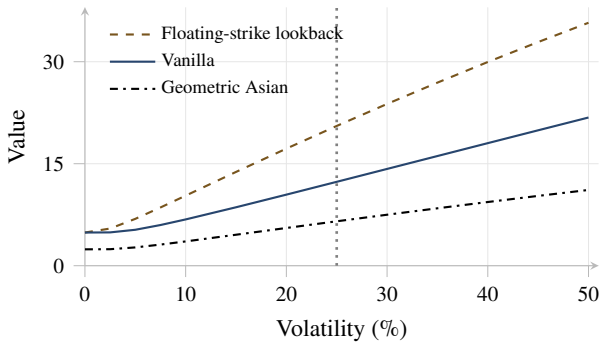


Figure 34.5: At-the-money calls against volatility: $S_0 = K = 100$, $r = 5\%$, $T = 1$, no dividends. The dotted line marks Example 34.7's 25%.

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34.10 BASKETS, RAINBOWS, AND EXCHANGE OPTIONS

Multiple underlyings are the second source of genuine difficulty, and their new input is correlation.

Example 34.8 What correlation is worth

Two assets at 100 with volatilities 25% and 30%, $T = 1$, $r = 5\%$, no dividends.

A Margrabe exchange option, the right to give up asset 2 and receive asset 1, has the closed form derived in Section 28.11 as equation (28.14):

$$V_0 = S_1 e^{-q_1 T} N(d_1) - S_2 e^{-q_2 T} N(d_2), \qquad \hat{\sigma} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \tag{34.17}$$

with

$$d_1 = \frac{\ln\left(S_1 e^{-q_1 T} / S_2 e^{-q_2 T}\right) + \hat{\sigma}^2 T/2}{\hat{\sigma}\sqrt{T}}, \qquad d_2 = d_1 - \hat{\sigma}\sqrt{T}.$$

Correlation enters only through $\hat{\sigma}$, so the table below is a table of Black–Scholes–Merton prices at five volatilities:

ρ	Effective volatility	Value
−0.50	47.70%	18.8495
0.00	39.05%	15.4808
+0.50	27.84%	11.0703
+0.90	13.23%	5.2737
+0.99	6.32%	2.5227

The value falls by a factor of seven across the correlation range. No interest rate appears, for the reason Section 28.11 gives: in units of asset 2 there is nothing to finance.

Now compare a *basket call* on $(S_1 + S_2)/2$ struck at 100 with a *basket of calls*, $(c_1 + c_2)/2$, half a 100-strike call on each asset:

ρ	Basket call	Basket of calls	Ratio
−0.50	8.3481	13.2836	0.628
0.00	10.3260	13.2836	0.777
+0.50	11.9088	13.2836	0.896
+0.90	13.0178	13.2836	0.980
+1.00	13.2792	13.2836	1.000

An option on a basket is always cheaper than a basket of options, with equality only at perfect correlation and equal volatilities, which is why the last row still shows a gap of 0.0044. The gap is the diversification the basket option does not let the holder keep: an option on each asset pays on whichever rises, while an option on the average pays only if the average rises.

Unlike the exchange option, the basket has no closed form, for the reason the arithmetic Asian does not: a weighted sum of lognormals is not lognormal. The workhorse approximation matches the first two moments of the basket $B = \sum_i w_i S_i$, with w_i units of asset i, to a lognormal,

$$M_1 = \sum_i w_i F_i, \qquad M_2 = \sum_i \sum_j w_i w_j F_i F_j \, e^{\rho_{ij}\sigma_i\sigma_j T}, \qquad \sigma_B = \sqrt{\frac{1}{T}\ln\frac{M_2}{M_1^2}} \tag{34.18}$$

where $F_i = S_i e^{(r-q_i)T}$, and then prices with Black's formula at forward M_1 and volatility σ_B. The approximation degrades exactly where the option is most interesting: against the exact values in the table it overstates by 0.06% at $\rho = 0.9$ and by 0.75% at $\rho = -0.5$, because a negatively correlated basket is the least lognormal. The geometric basket $\prod_i S_i^{w_i}$, by contrast, is exactly lognormal and has a closed form, which makes it the natural control variate here just as (34.15) is for the Asian.

The gap between a basket option and a basket of options is what Section 22.11's dispersion trade, short the index option and long options on its constituents, captures: it earns the difference between implied and realized correlation, and the ratio column shows how much is at stake.

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The modeling difficulty is that ρ has to come from somewhere. With two assets and a liquid option on each plus a liquid basket option, it can be implied. With fifty assets it cannot, because one basket price cannot determine 1,225 pairwise correlations, and the standard device (assume a single correlation for all pairs) is a one-factor approximation whose error is exactly Section 32.9's factor-reduction problem in a different market.

34.11 WORST-OF OPTIONS

One multi-asset payoff is among the most common exotics sold to retail investors, and its correlation exposure runs the opposite way to a basket's. With two assets it has a closed form and the derivation is one identity. Since $\min(S_1, S_2) = S_1 - \max(S_1 - S_2, 0)$, a forward on the minimum is a long position in asset 1 against a short exchange option (34.17), and the strike-K put on the minimum follows from

$$p_{\min}(K) = Ke^{-rT} - \underbrace{\left[S_1 e^{-q_1 T} - V_0^{\text{exch}}\right]}_{\text{PV of receiving } \min(S_1,S_2)} + c_{\min}(K) \tag{34.19}$$

Stulz's formula supplies the last term in bivariate normal probabilities:

$$\begin{aligned} c_{\min}(K) = {} & S_1 e^{-q_1 T} M(y_1,\, -d_1;\, -\rho_1) + S_2 e^{-q_2 T} M(y_2,\, d_2;\, -\rho_2) \\ & - Ke^{-rT} M\left(y_1 - \sigma_1\sqrt{T},\; y_2 - \sigma_2\sqrt{T};\; \rho\right) \end{aligned} \tag{34.20}$$

where $\hat{\sigma}$, d_1, and d_2 are the exchange option's (34.17), and

$$y_i = \frac{\ln\left(S_i e^{-q_i T}/Ke^{-rT}\right) + \sigma_i^2 T/2}{\sigma_i\sqrt{T}}, \qquad \rho_1 = \frac{\sigma_1 - \rho\sigma_2}{\hat{\sigma}}, \qquad \rho_2 = \frac{\sigma_2 - \rho\sigma_1}{\hat{\sigma}}$$

The exchange option appears twice, once as an instrument in (34.19) and once as the correlation structure inside M, which is the sense in which every two-asset payoff in this chapter is built from Margrabe's option.

Three assets is where the formula stops being usable. The same construction extends to n assets (Johnson's formula), but the bivariate M becomes an n-variate normal distribution function, and quadrature for those is already slow in three dimensions and unreliable above four. The three-asset worst-of that dominates the retail market is therefore simulated, which is why Example 34.9 reports a table rather than a formula, and why the issuer's risk numbers are Monte Carlo numbers with all the noise that implies.

Example 34.9 Why the coupon is high

Three assets, each at 100 with 28% volatility, $T = 1$, $r = 5\%$. A **worst-of put** struck at 100 pays on the *minimum* of the three. The benchmark is a vanilla put on any one of them, also struck at 100, which is worth 8.5955. The worst-of values are simulated, with 4 million antithetic pairs at each correlation; every standard error is below 0.005.

ρ	$E[\min S_T]$	Worst-of put	Relative to single-asset put
0.20	83.86	17.65	**2.05**
0.40	86.58	16.26	1.89
0.60	89.87	14.71	1.71
0.80	94.23	12.80	1.49
0.95	99.62	10.64	1.24

The worst-of put is worth more as correlation falls, exactly the opposite of the basket in Example 34.8.

A basket averages, so low correlation *reduces* dispersion and cheapens the option. A worst-of

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takes a minimum, so low correlation *increases* the chance that at least one asset falls a long way. At a correlation of 0.20 the expected minimum at one year is 83.86 against a starting level of 100—the drift is upward and the minimum still falls sixteen percent, purely from taking the worst of three.

A **worst-of note** is a bond whose principal is repaid in full only if the worst of the three assets finishes above the strike; otherwise the investor bears that asset's loss. The investor has therefore sold the issuer a worst-of put, and the issuer pays for it as a coupon above its own funding rate. At $\rho = 0.20$ the put is worth 17.65 per 100 of notional, about 18.5 percentage points of coupon on a one-year note once carried to maturity, and twice what a note on any single one of the assets could pay.

So the investor in such a note is long correlation, losing as it falls: the lower the correlation, the more the put is worth and the more the investor has given away. In a crash correlation rises, which on its own would lower the put's value, but it rises because everything is falling together, and with all three assets below the strike the put is deep in the money whatever the correlation. The investor therefore carries two losses for one coupon: the one that low correlation pays for, a single asset collapsing while the others hold up, and the market-wide fall, against which the rise in correlation offers no protection.

Adding assets to a worst-of always raises the coupon and always raises the risk, and a note advertised as "linked to three blue-chip stocks" is riskier than one linked to any of them individually. Section 35.8 returns to this as an instance of the gap between how these products are described and what they are.

Figure 34.6 puts this example and Example 34.8 on one axis. Both ratios come to one at perfect correlation, where the assets move as one, and they leave it in opposite directions as correlation falls: the basket call loses value against its components, while the worst-of put gains it. A worst-of note's investor is short the upper curve, which is why the coupon is highest exactly where the assets are least alike.

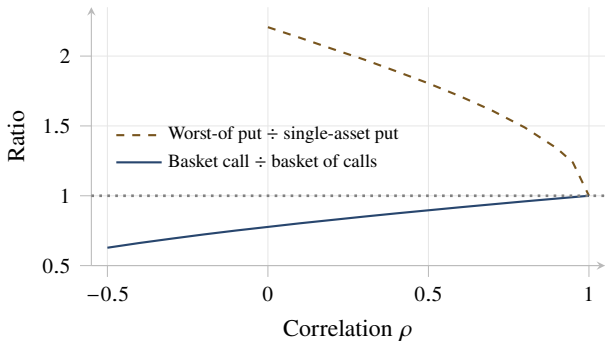


Figure 34.6: Examples 34.8 and 34.9 across correlation. The worst-of values are simulated.

34.12 CLIQUETS, AND THE CAP THAT REVERSES THE SIGN

Section 34.2 called a cliquet the purest instrument for trading forward volatility. The version actually sold is not, and Example 34.10 shows why.

Example 34.10 What the cap does

A three-year annual-reset cliquet paying $\sum_i \max(r_i, 0)$, where r_i is the year's return, each period capped at 8%. Under Black–Scholes each period is a forward-start at-the-money call spread, so its value depends only on that year's forward volatility. By the scaling in (34.9), the period payoff

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$\min\big(\max(r_i, 0),\ C\big)$ is worth, at its own reset date,

$$V = c(1,\ 1,\ \tau) - c(1,\ 1 + C,\ \tau) \tag{34.21}$$

per unit of notional, where $c(S, K, T)$ is the Black–Scholes–Merton call of Section 34.2, evaluated here at a spot of 1 and the period's forward volatility, $C = 8\%$ is the cap, and $\tau = t_i - t_{i-1}$ is the length of a period, one year here. That is a number, not a function of the level, so the whole contract is $\sum_i e^{-rt_{i-1}} V$. The table reports the undiscounted sum $3V$, which isolates the volatility comparison; discounting the second and third periods scales the total by about 0.95.

Forward volatility	Per period	Three-year cliquet
15%	0.03800	0.11401
20%	0.03670	0.11009
25%	0.03561	0.10682
30%	0.03463	0.10388
35%	0.03372	0.10115

The value falls as volatility rises. An uncapped cliquet would be long volatility; the cap makes this one short it, because at a 5% rate the one-year forward is 1.05, close to the cap at 1.08, and higher volatility helps the short leg of the spread more than the long one. At zero rates the same contract is long volatility up to about 30%: the sign is set by where the cap sits against the forward, not by the cap alone. That reversal is what lets it be sold at an attractive headline rate: it is marketed on upside participation and is, in volatility terms, a *short*-volatility position.

The model risk is severe, for a reason Section 21.3 supplies. Most of the value sits in the forward smile: the second and third periods are forward-starting, so what matters is the smile prevailing a year and two years from now. Section 21.3 measured what local volatility predicts. The skew of its local-volatility function (the slope of volatility against log-moneyness at the money) decays from -0.981 at three months to -0.231 at three years, so the model expects the smile a few years out to be far flatter than today's, while the market's forward smile is roughly stationary. A local-volatility model prices the later periods off a smile it has flattened, and a cliquet's value is concentrated exactly there.

This is the clearest case in the chapter where calibrating to today's surface is not merely insufficient but actively misleading, because the model reproduces every vanilla and gets the one quantity the product depends on structurally wrong.

Figure 34.7 shows the reversal against the contract the cap was applied to. The two slope in opposite directions, and an investor told they are buying participation in rising markets is holding the line that falls when volatility arrives.

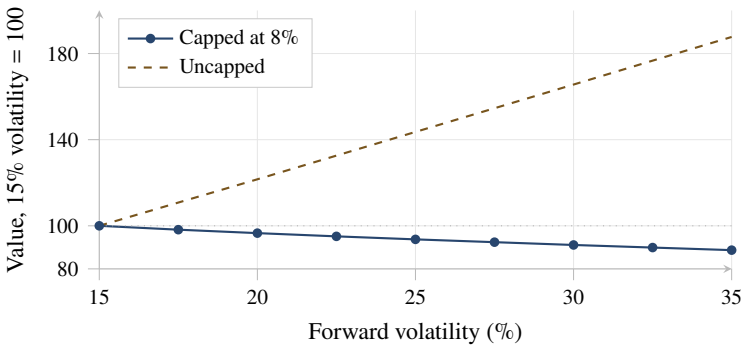


Figure 34.7: Three-year annual-reset cliquet of Example 34.10, each indexed to its own value at 15% volatility. In levels the capped contract falls from 0.11401 to 0.10115.

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34.13 WHAT THEY HAVE IN COMMON

The chapter's material sorts on one question, put as a test rather than a list. Can the payoff be replicated by traded instruments?

If it can be replicated statically (a package, a chooser, a digital as a limit of call spreads), then the vanilla market determines the price, model risk is small, and the residual is a hedging cost rather than a valuation uncertainty.

If it can only be replicated dynamically, then the price depends on the assumed dynamics, and calibration to vanillas constrains but does not determine it. Barriers, cliquets, lookbacks, and everything multi-asset live here. So does the arithmetic Asian, at the mild end: an average smooths the path so much that the dynamics barely move its price. Dynamic replication says where model risk can live; the payoff says how much of it there is.

The test is more useful than the taxonomy of Section 34.1 because it predicts the right thing. A chooser sounds as exotic as a compound and is two vanillas, while the compound's value at t_1 depends on an implied volatility that no surface quotes today; a forward-start option sounds trivial and carries as much model risk as anything in the book. It is also a different question from whether a formula exists, and the table sets the two side by side.

Instrument	Closed form under Black–Scholes	Model risk
Package, chooser	Exact, (34.8)	None
Compound	(34.7), plus one root search	Moderate
Digital	(34.13)	Low; the cost is hedging
Barrier, continuous	(34.11), all eight	**Severe**
Barrier, discrete	(34.11) with (34.12)	**Severe**
Lookback	(34.16)	High
Asian, geometric	(34.15)	Low
Asian, arithmetic	**None**; moment match or simulate	Low
Exchange	(34.17)	Correlation only
Two-asset max/min	(34.20)	Correlation
Basket; worst-of ($n \geq 3$)	**None**; simulate	Correlation, unhedgeable
Forward-start, cliquet	(34.9), (34.21)	**Severe**

The two columns are almost uncorrelated. Barriers, lookbacks, and cliquets all have closed forms and carry the worst model risk in the chapter; the arithmetic Asian has no formula at all and carries almost none. A closed form is a statement about the Black–Scholes–Merton model, not about the market: it returns the same number every time it is called, which is precision, not accuracy. The formulas are worth having because they are fast, exact within the model, and show where the model enters; they are not evidence that the price is known.

The test also explains a fact of market structure: the exotics with static hedges trade at narrower spreads than those without, because the dealer is charging for a known hedging cost rather than for a range of model prices.

Case Study 34.1 The structure that sold a discount and bought a multiple

The **accumulator** was sold across private banks in Hong Kong and Singapore through 2007 and into 2008, and it is this chapter's mechanism assembled into a single retail contract.

The terms sound like a discount. The client agrees to buy a fixed quantity of a stock on every trading day for a year, at a strike set a few percent below the spot price at inception. Nothing is paid at the outset. Two clauses complete it: the contract *knocks out* if the stock trades a few percent above the initial spot, and the daily quantity *doubles* if the stock is below the strike.

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Decomposed, it is three of this chapter's instruments: a strip of daily forwards at a discount; a short position in a strip of up-and-in forwards, which is what the knock-out clause amounts to, since it cancels the remaining forwards once the barrier trades; and a short position in a second strip of up-and-out puts at the same strike, which is what the doubling clause is. The discount is the premium of the options the client has sold, and the client is short volatility, short the downside at twice the size, and capped on the upside at a few percent.

What happened is what the structure specifies. Hong Kong and mainland equities peaked in October 2007 and fell through 2008. Because prices fell, the knock-out never triggered, so the contracts ran their full term rather than terminating early as they did in the case the seller described. The doubling clause then required the client to buy twice the daily quantity, every day, at a strike far above the market, with the obligation growing precisely as the losses grew. The instrument acquired a nickname that was a pun on its own name: *I kill you later*.

The failure was not mathematical. Nothing here needed a model the seller did not have. The knock-out was disclosed, the doubling was disclosed, and both behaved exactly as written. What was not conveyed was the asymmetry the two clauses create together—a payoff described by its best state and sized by its worst.

Three lessons connect it to the rest of the book. A barrier that removes the upside is worth more than it looks when it is one of two clauses rather than the only one. A zero-premium structure has a premium; Chapter 35's treatment of a note's estimated initial value gives the machinery for finding it, and here it is the discount itself. And leverage written into a contract's quantity is harder to see than leverage written into its notional, because nothing in the term sheet is labeled with a multiple.

34.14 WHO TRADES EXOTICS, AND WHY

Corporate hedgers are the economically motivated end of the market. An Asian option matches an averaged exposure and a barrier matches a hedge that is only wanted in some states: an importer who needs FX protection but not if the currency has already collapsed past the point where the business is unviable. These are exotics bought because the vanilla is the wrong shape. Basket options are bought by multinational treasurers hedging revenue in several currencies, for whom only the net matters, and an exchange option is what an outperformance option on one index against another amounts to.

Structured-note issuers supply the volume. Chapter 35 is about them, and almost every payoff in this chapter appears inside a retail note: barriers as the knock-in that ends the buyer's protection, digitals as coupon triggers, worst-of and best-of payoffs on several underlyings, cliquets as annual ratchets.

Dealers intermediate, and the pattern is one-directional. The client is almost always short the exotic optionality and the dealer long it, because the exotic feature is what cheapens the note or the hedge. So dealers run a structural long-exotic, short-vanilla book, which they hedge in the vanilla market, and the imprint Section 19.8 traces on the surface and Section 22.11's implied correlation both reflect that flow.

Hedge funds take the other side selectively, and the trades named in this book (dispersion in Section 22.11, forward-volatility positions against cliquets) are mostly bets that the dealer's hedging flow has pushed a price away from fair value.

The recurring structure is worth naming once. An exotic option transfers a risk the client cannot price or hedge to a dealer who can, and the dealer's compensation is the difference between what the exotic can be hedged for and what it was sold for. Sections 34.5 and 34.13

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determine how large that difference has to be.

SUMMARY

Exotics divide by path dependence, multiple underlyings, and non-standard exercise, but the useful division asks whether the payoff can be replicated by traded instruments. Statically replicable payoffs are priced by the vanilla market with small model risk; dynamically replicable ones depend on assumed dynamics, which calibration to vanillas constrains without determining.

A down-and-out call is worth well below the vanilla, and a reverse knock-out loses value as the underlying rises toward its barrier. Barriers are hard to compute as well as to hedge: a binomial tree's value jumps erratically with the number of steps unless a node layer sits on the barrier, and correcting for daily rather than continuous monitoring moves the price materially. The reflection hedge is exact only when $r = q$ and errs badly when they differ, and two models calibrated to the same vanilla surface priced one barrier far apart.

A digital is the limit of call spreads whose value converges while the position diverges, so dealers overhedge with a spread that dominates the payoff, at a modest premium over the digital. Every discontinuous payoff with a vanilla market beneath it is priced this way.

A lookback is worth substantially more than the vanilla and a geometric Asian substantially less: an Asian is a vanilla on a less volatile quantity, and a lookback removes the timing decision.

Correlation enters every multi-asset payoff and is rarely quoted. An option on a basket is cheaper than a basket of options, because the average gives up the dispersion the separate options keep. A worst-of put on several assets is worth far more than a single-asset put, because a minimum rewards dispersion where a basket's average punishes it, and that premium is what pays a worst-of note's coupon. A cliquet capped near the forward is short forward volatility, losing value as volatility rises, and its value sits in the forward smile, which today's surface does not pin down.

A closed form is a statement about the Black–Scholes–Merton model, not about the market. Barriers, lookbacks and cliquets have exact formulas and the worst model risk in the chapter; the arithmetic Asian has no formula and almost none.

Corporate hedgers buy exotics because the vanilla is the wrong shape, and structured-note issuers supply the volume. Clients are almost always short the exotic optionality, so dealers run a long-exotic, short-vanilla book hedged in the vanilla market.

FURTHER READING

Zhang's *Exotic Options* and Haug's *Complete Guide to Option Pricing Formulas* are the reference works for closed forms; Haug states each formula with a worked number, which makes it the place to check an implementation against, and both should be read knowing that a closed form exists only under assumptions the market violates. Derman, Ergener and Kani on static replication of barriers is the paper behind Section 34.5 and develops the strip-of-maturities refinement this chapter only mentions. Carr and Chou extend the reflection argument considerably further. Boyle and Lau (1994) is the source of Example 34.2's node-placement rule. Broadie, Glasserman and Kou on discrete barrier monitoring supplies the continuity correction of Section 34.6. On the multi-asset material, Margrabe's original paper is short and is the source of the exchange option; Stulz (1982) is the source of (34.20), and states put prices through the identity (34.19); and Johnson (1987) extends the result to n assets. The literature on basket approximations is large, and its methods differ mainly in accuracy, so decide what accuracy is needed before sampling it. For the model-risk question of Section 34.7, the practitioner literature on barrier reserves is thin and mostly internal; the academic treatment under uncertain volatility, starting from Avellaneda, Levy and Parás (1995), is the public

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substitute.

PRACTICE QUESTIONS

34.1 State the replication test that organizes this chapter, and use it to explain why a chooser carries less model risk than a forward-start option.

34.2 Using Example 34.1, state the fraction of the vanilla premium the barrier removes, and explain why knock-out deltas can exceed one.

34.3 Using Example 34.2, state the tree prices at 50 and 51 steps, explain the difference, and state the general lesson in a form that does not mention barriers.

34.4 Explain why grid-averaging fails to reduce the barrier tree error and why Richardson extrapolation makes it worse.

34.5 Derive the step count that places a node layer on the barrier, and state the error it achieves against an unaligned grid with more steps.

34.6 Using Example 34.3, state the static hedge error when $r = 5\%$ and $q = 0$, and explain why the condition for exactness is zero drift rather than zero rates.

34.7 Explain why a static hedge that is wrong by a known amount can be preferable to a dynamic hedge that is right in a model.

34.8 Explain why two models calibrated to the same vanilla surface can price a barrier 49.5% apart, and state the three defensible responses.

34.9 Using Example 34.5, state the cost of the $[95, 100]$ overhedge relative to the digital, and explain why the replicating call spread's notional diverges while its value converges.

34.10 Explain why an Asian option is worth about half a vanilla and a lookback about 1.67 times it, and why Asians dominate commercial usage while lookbacks do not.

34.11 Using Example 34.4, state the value of daily and of monthly monitoring relative to continuous, and explain why an unaligned tree that omits the continuity correction can land near the right price for the wrong reason.

34.12 Using Example 34.9, state the worst-of put's value relative to a single-asset put at $\rho = 0.20$, and explain why the correlation dependence runs opposite to a basket's.

34.13 Using Example 34.8, state the basket call's value relative to the basket of calls at $\rho = -0.5$ and at $\rho = 1$, explain what the gap represents, and explain why fifty assets cannot be handled with a single correlation.

34.14 Explain why the capped cliquet of Example 34.10 is short volatility, why most of its value sits in the forward smile, and why that makes calibration to today's surface actively misleading rather than merely insufficient.

34.15 Explain why dealers run a structural long-exotic, short-vanilla book, and why exotics with static hedges trade at narrower spreads than others.

34.16 Using (34.11), price the down-and-in call of Example 34.1 from the single expression C, and explain why the up-and-in call with $H < K$ is worth the vanilla.

34.17 Name three instruments in this chapter with closed forms and severe model risk, and one with no closed form and very little. Explain why the two properties are nearly unrelated.

FURTHER QUESTIONS

34.18 Set $r = q$ in (34.11) and show that the down-and-in call's C is exactly K/H puts struck at H^2/K. Then show where the argument fails when $r \neq q$, and relate the failure to the

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exponents 2λ and $2\lambda - 2$.

34.19 Example 34.2 shows a thousandfold error reduction from node placement. Derive the effective barrier as a function of n, show that the resulting price error is first-order in the barrier displacement, and use that to explain the sawtooth's shape between jumps.

34.20 Derive the reflection result of Example 34.3 from the symmetry of the lognormal density under $r = q$. Then determine how the static hedge error scales with $(r - q)T$, and design a multi-maturity strip that reduces it, stating what it costs.

34.21 Section 34.7 reports a 49.5% model spread on a barrier. Propose a reserving policy: specify which models define the range, how the reserve is released over the trade's life, and how you would prevent the policy from being gamed by choosing convenient models.

34.22 Example 34.5 shows the overhedge premium rising with the spread's width. Derive the dealer's optimal width as a trade-off between the premium given up and the expected hedging loss, and state how it depends on the expected holding period.

34.23 Example 34.8 shows a basket option below a basket of options at all $\rho < 1$. Prove the inequality in general, determine the conditions for equality, and connect the result to the convexity argument of Section 11.3.

34.24 This chapter argues that spreads follow replicability rather than payoff complexity. Test the claim against the exotics in this chapter: rank them by replicability, rank them by the spreads you would expect, and identify any instrument where the two orderings disagree. Then explain the disagreement.

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Chapter 35
Structured Products and Autocallables

This is where most people meet derivatives. Many households that have never traded an option own a structured note, sold by a bank branch or a private banker rather than a broker.

Chapter 12 said structured products are packaged strategies and took two apart with their option prices assumed. Chapter 34 priced the components. This chapter puts them together and asks the question the previous two could not: what does the buyer actually own, and is the price fair? Section 35.10 answers with a number: a $1,000 note worth $979 the day it is sold, the $21 sitting inside the coupon rather than beside it.

Every instrument here is legitimate, and most serve a real purpose. The trouble is in how they are sold: on a headline number (a participation rate, a coupon), while the risk sits in terms the headline leaves out. That gap is the subject of this chapter.

Section 35.1 takes the common notes apart into a bond and options, and Sections 35.2 and 35.3 take up the two that dominate, the reverse convertible and the autocallable, price the autocallable, and show what its return distribution looks like. Sections 35.4 and 35.5 cover the secondary market and the remaining families. Sections 35.6 to 35.8 turn to the issuer's side: how the embedded option is hedged, what happens when many notes share a barrier, and why most notes reference the worst of several underlyings. Sections 35.9 to 35.11 are about what the buyer pays for (the issuer's credit and the fee) and give a procedure for evaluating a note; Section 35.12 asks what disclosure can and cannot fix, and Section 35.13 who buys these products and why.

35.1 TAKING A NOTE APART

Every structured note decomposes into a bond and a derivative. The decomposition is the most useful skill in this chapter, because it converts a marketing document into a portfolio.

The decompositions are short, and the rest of the chapter refers back to them. Write F for the face amount, S_0 and S_T for the underlying at issue and at maturity, and $N = F/S_0$ for the number of option units a note of face F carries. In every case the redemption is a bond plus a position in options, and the only question is which options and on which side.

Principal-protected note. The buyer is promised the full face back at maturity whatever the underlying does, plus a share p, the **participation rate**, of any rise in the underlying over the note's life. A note with 90% participation on an index that rises 20% repays 118 per 100 of face; if the index falls, it repays 100. At maturity the redemption is

$$V_T = F + p\,N\,(S_T - S_0)^+ \tag{35.1}$$

a zero-coupon bond plus pN calls struck at the initial level. The bond secures the guarantee, and whatever the bond leaves over buys the calls, which is why the participation rate is a residual rather than a promise, and why Example 35.1 finds it governed by the issuer's funding cost.

Capped participation. The same promise, except that the share of the rise is paid only up to a ceiling: above a level C the redemption stops growing. In exchange the participation below the cap is higher, often above 100%. Capping the upside at C turns the call into a call spread:

$$V_T = F + p\,N\left[(S_T - S_0)^+ - (S_T - C)^+\right] \tag{35.2}$$

The sold call is what pays for the higher headline participation. A note advertising a participation above 100% has either sold something, usually this call, or is paying for its issuer's credit, as Example 35.1 shows.

Reverse convertible. The buyer receives a fixed coupon c well above deposit rates. At maturity the face is repaid in cash if the underlying stock or index ends at or above a barrier H; if it ends below, the buyer receives shares, or their cash value, instead and so takes the whole fall from the initial level:

$$V_T = F(1 + c) - N\,(S_0 - S_T)^+\,\mathbf{1}_{\{S_T < H\}} \tag{35.3}$$

a bond and a coupon, less a put the investor has written: struck at the initial level, but paying only when the underlying ends below H. Setting $H = S_0$ deletes the indicator and leaves a plain short put, which is the un-barriered version of the product. The indicator is also the discontinuity: the note pays $F(1 + c)$ just above H and $F(1 + c) - N(S_0 - H)$ just below it, and Section 35.6 is about what that does to the hedge.

Shark fin. Principal is protected and the buyer shares in the rise, usually at a high participation rate, but only as long as the underlying never trades above an upper level H during the note's life. If it ever does, the participation is lost and the buyer receives a small fixed rebate R instead. The name comes from the shape of the payoff, drawn in the bottom-right panel of Figure 35.1: flat at par below the initial level, rising steeply along the participation line up to H, then dropping back to par plus the rebate, the outline of a shark's dorsal fin. At maturity

$$V_T = F + p\,N\,(S_T - S_0)^+\mathbf{1}_{\{\max_{t\le T} S_t < H\}} \;+\; R\,\mathbf{1}_{\{\max_{t\le T} S_t \ge H\}} \tag{35.4}$$

a bond, an up-and-out call, and the rebate. The knock-out is cheaper than the vanilla call of (35.1), so the same budget buys more of it: that, and not a better view, is where a shark fin's headline participation above 100% comes from. Shark fins come in two forms. Some observe the barrier only at maturity, so the payoff depends on S_T alone and Figure 35.1 draws it exactly; others observe it daily or continuously, which is what (35.4) writes and Example 35.4 prices, and then a path that touches H and falls back is knocked out too.

Autocallable. The buyer receives a high coupon, but the note can end early. On each of a series of observation dates, if the underlying is at or above a trigger level, the note is *called*: it repays the face plus all the coupons accrued so far, and stops. If it is never called, it repays the face with all the coupons at maturity unless the underlying has closed below a barrier on any day, in which case the buyer is repaid in shares, takes the loss as in a reverse convertible, and forfeits the coupons. In symbols, take observation dates $t_1 < \cdots < t_n = T$, an autocall trigger K_a, a coupon c per annum with each t_i measured in years, and a barrier H, observed daily, as in Example 35.2. Let τ be the index of the first observation date on which $S_{t_i} \ge K_a$, and $\tau = \infty$ if that never happens. Then

$$V = \sum_{i=1}^{n} F(1 + c\,t_i)\,\mathbf{1}_{\{\tau = i\}} \;+\; \left[F(1 + cT)\,\mathbf{1}_{\{\min_{t\le T} S_t \ge H\}} + NS_T\,\mathbf{1}_{\{\min_{t\le T} S_t < H\}}\right]\mathbf{1}_{\{\tau = \infty\}} \tag{35.5}$$

a strip of digital payments, each conditional on not having been called already, plus, if the note is never called, a written down-and-in put: struck at the initial level like that of (35.3), but triggered by any daily close below the barrier, after which the buyer receives shares worth NS_T and forfeits the coupons. Two things follow that the **term sheet** (the document setting

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out the note's terms for the buyer) does not say. *The investor is short volatility twice*: once in the digitals, which want the underlying quietly above K_a, and once in the put. And a reverse convertible is the case $n = 1$ with no trigger, except that it keeps its coupon below the barrier and observes the barrier only at maturity. The two share their downside, a written put that pays only once the barrier is breached, and are sold to the same buyer on the same high coupon, but their risk profiles differ: early redemption cuts the autocallable's expected life well below its stated term, so the buyer collects the coupon for less time while carrying the same tail, and the outcome depends on the path as well as on where the underlying ends.

Dual-currency deposit. The buyer places a deposit in one currency and earns a rate above the ordinary deposit rate. At maturity the bank chooses the currency of repayment (the original one, or a second currency at a conversion rate fixed at the outset), and it will choose whichever is worth less to the depositor. Write X_T for the exchange rate at maturity, in units of the second currency per unit of the deposit currency, K for the conversion rate, and c for the interest, paid in the deposit currency whatever happens. Measured in the deposit currency, the redemption is

$$V_T = F(1 + c) - \frac{F}{X_T}\,(X_T - K)^+ \tag{35.6}$$

a deposit less a written call on the deposit currency struck at the conversion rate. When the deposit currency strengthens past K, the principal comes back converted at K and is worth only FK/X_T. Example 35.5 carries the arithmetic, and the enhancement is the option premium, as it is in every structure above.

One identity organizes all of them. At issue,

$$\text{issue price} = \underbrace{\text{PV(principal)} + \text{PV(coupons)}}_{\text{the bond}} + \underbrace{\text{options bought} - \text{options sold}}_{\text{the derivative}} + \text{fee} \tag{35.7}$$

Figure 35.1 draws the first four on one scale, and the shapes carry the argument. In each panel the dashed line is what the same 100 would return left on deposit, at 4% for a year, so the gap between it and the solid line is what the embedded options add or take away against simply earning interest. The principal-protected note repays 100 where the deposit pays 104, which is the interest given up to buy the participation; the reverse convertible pays 110 above its barrier, the extra 6 being the premium for the put the buyer has written, and falls far below the deposit once the barrier is breached. Where each solid line crosses the dashed one marks the range of the underlying in which the buyer comes out ahead:

- above 104.4 for the principal-protected note;
- above 104 for the capped note;
- anywhere at or above the barrier of 70 for the reverse convertible;
- only between 103.6 and the knock-out at 130 for the shark fin.

Each note is a bet that the underlying ends in its own range, and the reverse convertible's range is the widest because its losses, when they come, are the largest. Every term but the fee is quotable, which makes the decomposition worth doing: price the bond at the issuer's own funding curve, price the options in the market, and the residual is what the note costs to own.

Example 35.1 Where the participation rate comes from

Example 12.7 took this note apart: five years, a zero-coupon bond worth \$79.85 per \$100 of face at 4.5% (\$798.52 on the \$1,000 face used here), the option budget it leaves, and what a 2% fee takes out of that budget. It assumed the call at 25% of notional and held that price fixed. Here the index yields 2.0% with 20% volatility and the market rate is 2.0%. The at-the-money call is bought in the market and costs \$160.10 whatever the issuer pays; only the bond is discounted at the

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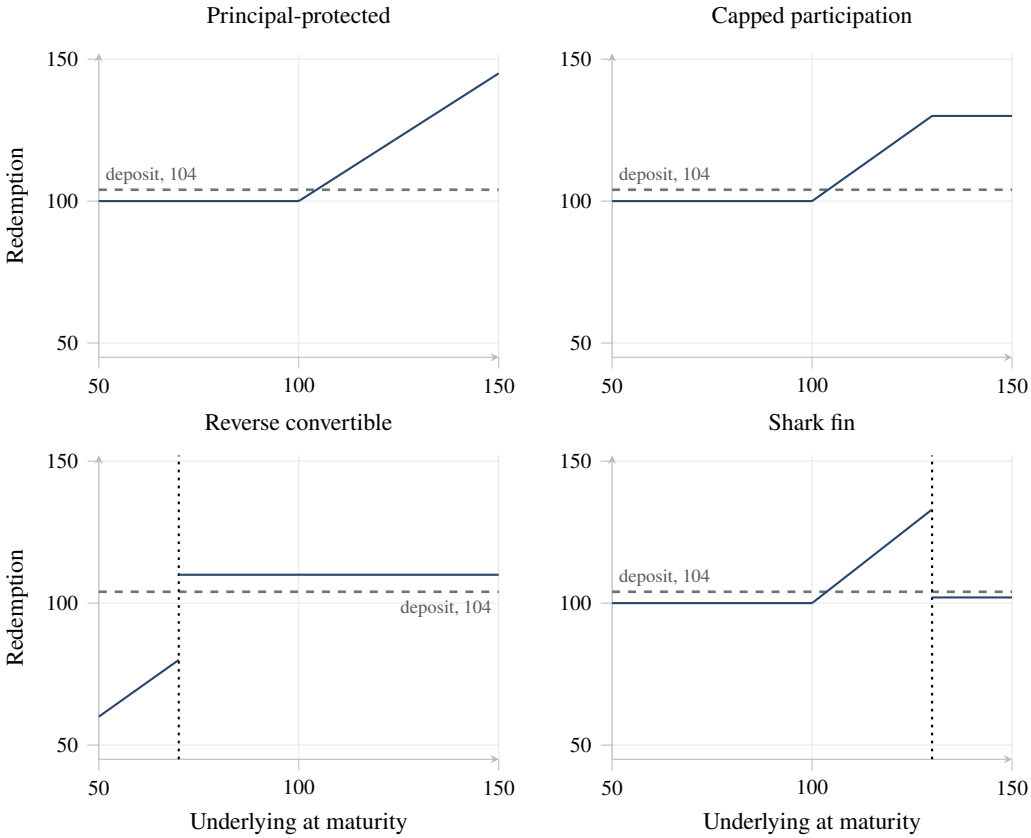


Figure 35.1: Redemption per 100 of face, initial level 100. Participation 90%; 100% to a cap of 130; barrier 70 with a 10% coupon; 110% to a knock-out at 130 with a 2% rebate. Dashed: 100 left on deposit at 4% for a year. Illustrative.

issuer's own funding rate, which is why the 4.5% row below reads 125.8% where Example 12.7's assumed price gave 80.6%:

Issuer funding rate	Participation (no fee)	(2% fee)
2.0%	59.4%	46.9%
3.0%	87.0%	74.5%
4.5%	125.8%	113.4%
6.0%	161.9%	149.4%

The participation rate is mostly a statement about the issuer's funding cost, not about the index. It nearly triples as funding goes from 2% to 6%, with the index unchanged.

The market rate is common to every issuer at one moment, so each row is a funding spread over it, from none to 4%, and a higher participation rate signals a weaker credit. The issuer paying more to fund can buy more optionality, so the best-looking note in a comparison is systematically the one with the most credit risk; Section 35.9 says what that has cost.

35.2 REVERSE CONVERTIBLES AND AUTOCALLABLES

The reverse convertible comes first, because an autocallable is a reverse convertible with early redemption added. The investor in a reverse convertible is short the put of (35.3), and the coupon above the deposit rate is the premium, as Example 12.8 showed for a 12% coupon. A

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note paying 10% when deposits pay 4% is not offering 6% of extra yield; it is paying 6% for a put the investor has written. The mis-selling risk is structural rather than incidental. The product is described in the vocabulary of fixed income (coupon, maturity, principal), and its market risk is entirely equity, so a buyer who compares it with a bond is comparing the wrong thing.

The autocallable is the dominant structured product globally, and the rest of this chapter is built on it. The mechanics are simple to state and hard to price. The note pays a coupon and terminates early ("autocalls") if the underlying is above a level on an observation date. If it never autocalls, principal is returned at maturity together with the accrued coupons *unless* the underlying has breached a lower knock-in barrier, in which case the investor takes any fall below the initial level and forfeits the coupons.

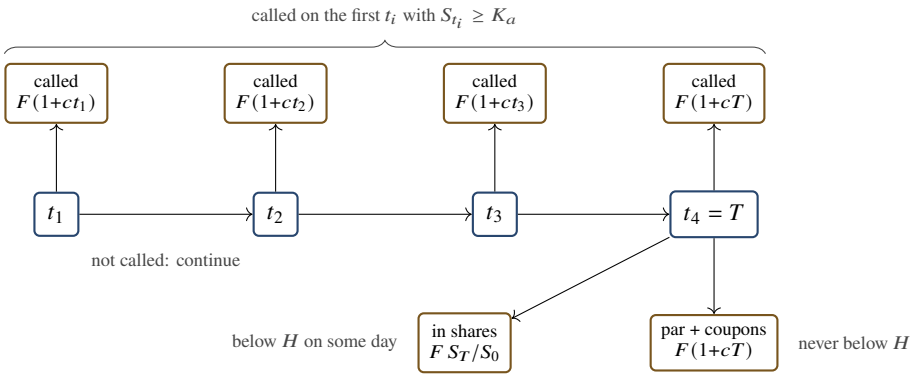


Figure 35.2: Each observation either redeems the note with the coupons accumulated to that point or passes it on. Only the final date can repay in shares.

Figure 35.2 is (35.5) drawn. The note is a sequence of chances to be redeemed early, each paying the coupons banked so far, and only the last one can end in shares.

Example 35.2 Pricing one, and what the buyer owns

A two-year note on an index at 100. Autocall at 100% observed monthly after a two-month **lockout period**, during which the note cannot be called, so the first observation is at month three; knock-in at 75% observed daily. The coupon accrues at an annual rate c: a note called at the end of month m repays $1 + c\,m/12$, and one that is never called and never knocked in repays $1 + 2c$ at maturity. If knocked in and never autocalled, the investor receives $\min(S_T/100, 1)$ per unit of face and no coupon. Volatility 25%, rate 2.5%; 80 million antithetic paths (200 seeds of 400,000), which put the fair coupon's standard error at 0.005 of a percentage point. The coupon c is the unknown: it is set so that the note is worth exactly its issue price of 1.00, which makes it the **fair coupon**.

P(autocalls at some point)	0.8311
P(knocks in before being called)	0.2153
P(**knocked in and never autocalled**)	**0.1645**
Fair coupon	**18.42% per annum**

At an 18.4% coupon the note prices at par. Nothing is being given away: the coupon is interest on the principal plus exactly what the embedded short put is worth. That holds at the one volatility assumed. A real desk prices the note on the volatility smile, and Section 21.11's case study shows that the choice of model for its evolution moves both the fair coupon and the hedge.

The probability structure is what sells it and what should worry the buyer. The note produces its advertised return about 83% of the time. That is why these products are easy to sell and why

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holders report satisfaction for years at a stretch. The remaining 17% is where all the loss is, and it is not a small loss—a knocked-in note pays $\min(S_T/100, 1)$, so the investor absorbs the full downside from par.

This is a negatively skewed payoff (frequent small gains, rare large losses) sold on its mean and its hit rate, which is the same structure as Section 22.8's inverse volatility products and Section 33.6's range accrual, and it recurs because it is what sells.

Now consider the exposure the buyer does not see:

Realized volatility	**Note value at** $c = 18.42\%$
15%	1.0520
20%	1.0204
25%	**1.0000**
30%	0.9847
40%	0.9574

The note is short volatility. It was priced at 25%; realized volatility above that makes it worth less than par, and the investor never agreed to a volatility view. It is short volatility in the worst way, because volatility rises when markets fall, which is exactly when the knock-in is approached. The two risks are not independent—this is Section 19.9's leverage effect and Section 27.9's wrong-way structure appearing in a retail product.

35.3 WHAT THE RETURN DISTRIBUTION LOOKS LIKE

A fair price says the mean is right. It says nothing about the shape, and the shape is what the buyer is actually choosing.

Example 35.3 The same fair price, a different distribution

Simulate Example 35.2's note at its fair 18.42% coupon on the same 80 million paths, and record the investor's total return, then do the same for simply holding the index for two years.

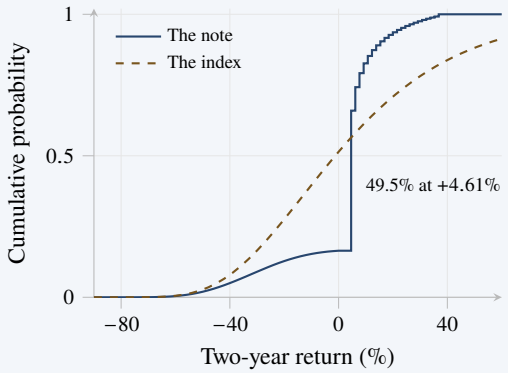
	The note	**The index**
Mean total return	+1.60%	+5.13%
Standard deviation	17.25%	38.36%
Skewness	**−1.59**	**+1.14**
Excess kurtosis	2.83	2.41
P(positive)	**83.55%**	48.59%
P(loss > 25%)	11.02%	21.82%
5th percentile	−40.03%	−44.79%

Everything here is under the risk-neutral measure, so both instruments earn the risk-free rate by construction. The note's lower mean is not a worse deal: it terminates early, after 0.70 years on average, so it earns that rate over a shorter life. That gives $2.5\% \times 0.70 = 1.75\%$, against the simulated 1.60%, which falls a little short because the paths that run longest, and so carry the largest growth factor, are the knocked-in ones that lose; that is the internal check that the pricing is consistent.

With the mean neutralized, the comparison is purely about shape, and the shape is the product.

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Median return	+4.61%
E[return | positive]	+8.22%
E[return | negative]	**−32.05**%
Average of the worst 1%	−60.23%



Wins average 8% and losses average 32%, a four-to-one asymmetry at a five-to-one win rate; a fair price fixes only their probability-weighted average. Three observations follow.

Skewness measures a distribution's asymmetry: it is negative when the long tail lies on the loss side, so that small gains are frequent and large losses rare, and positive when the long tail lies on the gain side. (Excess kurtosis, the row below it, measures how much more of the distribution sits in the tails than a normal distribution's, for which it is zero.) The note's skewness is negative and the index's is positive. A buyer who moves from the index into the note has kept almost all of its tail and given up its upside; they have exchanged a positively skewed exposure for a negatively skewed one with both earning the risk-free rate, which is the opposite of what "capital-efficient" and "defensive" suggest.

Nearly half the distribution, 49.5%, is a single point (the note autocalled at the first opportunity), so every percentile from the 17th to the 65th, the 25th and the median included, is +4.61%, which is why holders experience these products as reliable. The plot beside the second table draws both distributions cumulatively: the note's single point is the 49.5-point vertical step at +4.61%. Below zero the note's curve lies under the index's, because a loss is far less likely (16.45% against 51.41%), but the note can never pay more than +36.84%, a return the index exceeds 18% of the time.

The standard deviation is genuinely lower than the index's, 17.25% against 38.36%, so a risk measure that sees only the second moment will report this product as less than half as risky. A tail measure reports it as slightly less risky, and no more: the worst 1% of outcomes averages −60.23% for the note and −61.29% for the index. What the note gave up for its lower variance is the index's upside, not its tail. Chapter 38's warning that a single-number risk measure can miss what sits in the tail is not abstract; this is the product it describes.

35.4 THE SECONDARY MARKET

A structured note is quoted daily and is not liquid, and the distinction matters more than it sounds.

The issuer is usually the only bid. There is no exchange, no competing market maker, and no obligation to quote. A buyer who wants out sells back to the party who sold it to them, at a price that party computes. Three consequences follow.

The bid embeds a second spread. Having paid the issue premium of Section 35.10, an early seller pays again on the way out. Round-tripping a note in the first year can cost several percent of principal before any market move.

The quoted price is a model output. These are Level 2 or Level 3 valuations in the accounting sense (prices computed from a model, with observable or unobservable inputs respectively, because no active market quotes the instrument), and Section 26.12's observation about the crisis applies in miniature: when the only price comes from the party on the other side, the holder cannot distinguish a mark from a quote.

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And illiquidity interacts with the payoff. An investor most wants to exit after a sharp fall, when the knock-in is close and the outlook has changed, and that is exactly when the bid is lowest and the embedded option's value has moved most against them. The exit is worst when it is most wanted, which is this book's wrong-way structure appearing again.

Case Study 35.1 A market that learned this from its losses

South Korea developed one of the world's largest retail markets for autocallable notes, sold as **equity-linked securities** through bank branches to ordinary savers. A large share referenced the Hang Seng China Enterprises Index (HSCEI), chosen because its high volatility supported high coupons.

The structure worked as designed for years. Notes autocalled, holders were repaid early with the advertised coupon, and the proceeds were rolled into new notes: the 83% outcome of Example 35.2, repeated often enough to look like a rule rather than a probability.

Then the index fell a long way and stayed there. Notes issued in 2021 carried knock-in barriers set against 2021 levels; by the time those notes reached maturity in 2024, the index had not recovered, and the loss that Example 35.2 gives a 17% probability became the realized outcome for a large cohort at once. Four features of the episode are general.

The losses were synchronized because the issuance was. Notes sold during one period share a vintage, a barrier level, and a maturity. A product with a 17% loss probability sold to a hundred thousand people in the same quarter does not produce 17% of them losing independently; it produces nearly all or none.

The distribution channel mattered as much as the product. The notes were sold through bank branches to savers, alongside deposits, by staff whose training and incentives were not designed for a negatively skewed derivative.

The regulatory response focused on sales practice rather than on the instrument: **suitability** assessment (a check, before the sale, that a product fits the buyer's circumstances, knowledge and objectives), recording requirements, limits on where such products may be sold. That is the right diagnosis if one accepts that the instrument is legitimate and the mismatch was between the payoff and the buyer. The supervisor then turned to enforcement: in November 2025 it gave notice of about 2 trillion won of fines on five banks for mis-selling the HSCEI notes, which the Financial Services Commission finally set in November 2026 at about 360 billion won.

The mechanism was entirely visible in advance. Everything in Sections 35.2 through 35.3 could have been computed at issue from the term sheet. Nothing about this episode required hindsight.

35.5 OTHER FAMILIES

Four more structures account for most of the rest of the market, and each is a recombination of what has already been priced; a fifth, the hybrid, is not.

Shark-fin notes, Section 35.1's up-and-out call plus a rebate, sell the tail of the upside, and the appeal is that giving up the potential of returns above 30% feels costless to someone who expects 10%. Example 35.4 prices what the feeling costs.

Example 35.4 Where the headline participation comes from

A one-year \$1,000 note at a 3% rate, on an index at 100 with 25% volatility. The zero-coupon bond costs 970.45, leaving **29.55** for the option and its rebate.

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An at-the-money call is worth 11.3485 per 100 of index, so that budget buys **26.0%** participation, an unattractive number to print on a term sheet. Adding a knock-out at 130 drops the option to 2.1766, but a shark fin also pays a rebate, here 2% of face at maturity if the barrier is touched. The touch probability, from Section 34.3's formula, is 0.2924, so the rebate costs $1{,}000 \times 2\% \times e^{-0.03} \times 0.2924 = 5.68$, and the remaining 23.88 buys

$$\frac{23.88}{10 \times 2.1766} = \textbf{109.7\% participation}$$

The participation rate has more than quadrupled, and what has been given up is every path that touches 130 at any time before maturity, not only those that end above it.

The question the term sheet does not answer is how often those outcomes occur. That touch probability, **29.2%** under the risk-neutral measure, is the answer: in nearly a third of scenarios the 109.7% participation is extinguished and the note pays its small rebate—and the knock-out happens in the states where the index rallied hardest. The buyer has not sold a remote tail; they have sold roughly the three strongest rallies in ten.

Two features make this harder to see than the autocallable's. When the barrier is monitored continuously, as here, a single intraday touch during a rally destroys the participation for the rest of the year even if the index closes the year at 128. And the rebate is quoted as a percentage of face rather than as what it replaces: a 2% rebate against a forgone 109.7% participation on a 30% move is 2% against 32.9%. The instrument is Example 34.6's reverse knock-out sold retail, with the same discontinuity and none of the warning.

Dual-currency deposits, the deposit plus a short FX option of (35.6), are among the most widely sold structured products in Asian private banking. The presentation as a "deposit" is the problem: the principal is at risk in currency terms and the vocabulary says otherwise. Example 35.5 shows that the entire enhancement is the option premium, to the basis point.

Example 35.5 The deposit that is not one

A three-month dollar deposit of $100,000 with a conversion rate of 153 against a USD/JPY spot of 150. Dollar deposits pay 4.5%, yen 0.5%, and three-month volatility is 10%. The depositor has written a dollar call struck at 153: if the dollar strengthens past it, the bank repays in yen at 153. Interest is paid in dollars whatever happens; only the principal can be converted.

Chapter 16's Black–Scholes–Merton formula, with the yen rate for discounting and the dollar rate as the yield q (the form known as Garman–Kohlhagen), values that option at **1.2776** yen per dollar, which is 0.852% of the dollar notional today, 0.861% at maturity, or **3.45% annualized**. So:

$$4.5\% \text{ (deposit)} \;+\; 3.45\% \text{ (written option)} \;=\; \textbf{7.95\%}$$

The advertised 7.95% rate is the deposit rate plus the premium, and nothing else. A depositor who understood the decomposition would ask whether 3.45% a year is a fair price for the option—which is the question the product's presentation prevents, because a deposit rate is not something one negotiates.

The outcomes are asymmetric in the familiar direction:

USD/JPY at maturity	Repaid in	Quarter's return
140	dollars	+1.99%
150	dollars	+1.99%
153	dollars	+1.99%
160	yen at 153	−2.39%
165	yen at 153	−5.29%
175	yen at 153	−10.59%

The gain is capped at one quarter's coupon and the loss is not capped at all, which is Section 35.2's shape in a three-month product and a different asset class. And the conversion

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arrives at the worst moment: a dollar-based investor is repaid in yen precisely when the yen is weak.

Credit-linked notes embed the short CDS position of Section 25.15. The whole of Part V's credit machinery applies, and the retail versions are the ones where the reference entity's identity is least understood by the buyer.

Range accruals appear as notes as well as swaps, with the coupon arithmetic and the short-vega exposure of Section 33.6 unchanged.

Hybrids attach a second condition from another asset class to the credit one (a coupon paid only while an issuer survives *and* an index is above its strike), and they are the exception to this section's arithmetic. The two legs cannot be priced separately and added, because the value depends on a correlation between default and the other factor that neither market quotes; Section 25.16 works the example, and the practical consequence for a buyer is that the usual decomposition is unavailable to them as well as to the dealer.

The pattern across the first four is the one this chapter began with. The enhanced headline term is paid for by an option the buyer has written, and the option is described by its trigger rather than by its risk.

The buffer ETF of Section 12.10 is the contrast case. It holds a deep in-the-money call, a put spread that supplies the buffer, and a sold call that finances the cap, and it resets each year. Its cap and buffer are quoted, its terms are rules rather than a negotiated term sheet, no issuer's credit is involved, and its cost shows in the expense ratio, so it supplies the comparison for Sections 35.9 and 35.10.

35.6 ISSUER HEDGING, AND THE DISCONTINUITY

The investor's position implies the issuer's, and the issuer's hedging is where these products stop being a private matter between two parties. The investor is short a down-and-in put; the issuer is long it. A long put has negative delta, so the issuer hedges by holding the index, and the size of that holding is what matters.

Example 35.6 Why the hedge stabilizes, until it does not

The embedded option is a down-and-in put struck at 100 with a barrier at 75, one year, with Example 35.2's 25% volatility and 2.5% rate and no dividends, valued exactly with the down-and-in put row of Section 34.3's barrier formulas. The note's knock-in is observed daily, which Section 34.6's correction handles by moving the barrier away from the spot, to 74.32; that lowers the value at 100 from 6.160 to 5.956 and leaves the shape of the delta, and its jump at the barrier, essentially unchanged, so the continuous barrier is used below.

Index	Option value	Delta
130	0.706	−0.055
110	3.128	−0.220
100	6.160	−0.399
90	11.365	−0.652
80	19.262	−0.921
76	23.124	−1.006
75.33	23.802	**−1.017**
74.67	24.411	**−0.827**
74	24.968	−0.836
70	28.414	−0.885

Above the barrier, delta steepens monotonically from −0.055 to −1.017 as the market falls, and it *exceeds one in magnitude*, a barrier feature Section 34.3 flagged.

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The issuer, long this option, must therefore hold a growing long position in the index as the market falls. From 130 down to the barrier the hedge grows from 5.5% to 101.7% of notional. The issuer is buying into a falling market, continuously, and that flow is stabilizing. It is the gamma term of Chapter 18's expansion, equation (18.1), with the sign Section 18.12 showed dampens a move: above the barrier the issuer is long gamma, so staying delta-neutral means buying as the market falls.

Then the barrier is touched and the delta jumps from −1.017 to −0.827: a change of +0.19.

Two things happen at once. The issuer must *sell* about 19% of notional immediately. And its bid weakens: below the barrier the option is an ordinary put, and the issuer's buying, which ran at about 2.8% of notional per index point at 90 and 1.6% just above the barrier, drops to 1.4% just below it and 1.1% by 70.

The popular account of these episodes emphasizes the forced selling, and for one note the arithmetic agrees: a further ten-point fall costs about 5% of notional in lost buying, against a 19% sale at the barrier. What the account misses is the support that came before it. The issuer had been buying all the way down, so what a regulator should monitor is not only the sell order at the barrier but how much stabilizing flow weakens at a given index level, and Section 35.7 shows that no single issuer can see that.

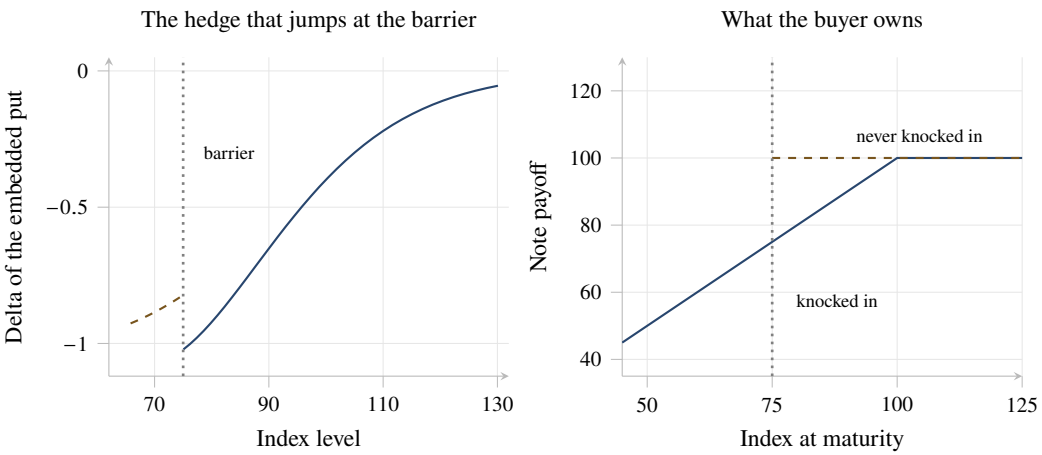


Figure 35.3: Left: Example 35.6's delta. Right: maturity payoff, coupon excluded.

Figure 35.3's left panel is the mechanism. The gap at the barrier is the discontinuity, and the slope on either side of it is the stabilizing flow: steepest around 90, and shallower below the barrier than above it. Its right panel is the maturity payoff, coupon excluded: flat at par if the barrier was never touched, and the index level itself, up to par, if it was.

35.7 WHEN MANY NOTES SHARE A BARRIER

One note's hedging is irrelevant to a market. The problem is that barriers cluster.

Notes are issued continuously against a rolling reference level, so a sustained decline of a given size brings a large cohort of barriers into range at once. The clustering is not coincidence but arithmetic: barriers are set at round fractions (75%, 80%) of the level prevailing at issue, so notes issued during any period of stable prices share nearly the same absolute barrier. Three consequences follow, in order of increasing seriousness.

The stabilizing flow is larger than any one issuer knows. Each dealer sees its own book, which is Section 27.13's aggregation problem in a different market.

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The sale is synchronized. When the index passes the cluster, many issuers sell and their bids weaken within the same few sessions.

The feedback is self-reinforcing while it lasts. The buying on the way down slows the decline, which keeps the index above the barriers longer, which sustains the buying—until it does not. A stabilizer that works until it stops is not a stabilizer, and Section 23.10's saturating pin risk made the identical point about a different product.

Case Study 35.2 A cluster passed through: China, January–February 2024

China's structured-product market is built around the **snowball**, the domestic form of the autocallable, and its standard terms track Example 35.2 closely: a CSI 500 or CSI 1000 underlying (the China Securities Index Company's mid-cap and small-cap indices), twelve to twenty-four months, monthly knock-out (autocall) observation after a lockout of about three months, a knock-in observed daily at 70–80%, and coupons of 15–20% a year. The underlying is chosen for its volatility, which is what pays the coupon (the same reason Korea's notes referenced an offshore China index), and that coupon range is the same order as the 18.42% of Example 35.2, because it is the same option at the same volatility.

One feature of the market differs from Korea's in the way that matters most. Snowballs are sold by securities firms acting as counterparty, either issued directly as the firm's own income certificates or wrapped in asset-management products with a minimum subscription around 1 million renminbi; either way the buyer is a qualified investor rather than a depositor. The second of the Korean episode's four general features, that the distribution channel matters as much as the payoff, had been addressed institutionally, and in advance.

What had not been addressed was the synchronization. Barriers were set at 75% or 80% of the index level prevailing at issue, so notes written through a comparatively stable 2023 shared nearly the same absolute knock-in level. The decline of January and February 2024 went through it.

Both of Example 35.6's effects are identifiable in what followed. Dealers hedging the long down-and-in put had been buying index futures as the market fell, supporting it. At the barrier they had to sell part of the hedge at once (about a fifth of notional for Example 35.6's note), and their buying slowed thereafter. So snowball hedging did not cause the fall; it first slowed the decline and then, at the barrier, added to it—and in a price series, "added to" looks a great deal like "caused." Attribution is harder still because a second leveraged chain, quantitative funds' equity positions financed through broker swaps, was unwinding in the same week, and separating the two needs dealer-level position data that is not public.

The regulatory response is the informative part, because most of it ran ahead of the event. Suitability requirements for snowball sales, an explicit instruction that the product is not principal-protected, and later limits on how much of a private fund could be put into snowballs (reported from 2022) and on how much a dealer could issue against its capital were all in place before 2024. None of them addressed the clustering, because suitability governs who may buy, and the cluster is a fact about how many investors bought notes with nearly the same barrier level in the same quarter. A rule aimed at individual prudence does not solve a problem about an aggregate, which is Section 27.13's conclusion, reached again in a different market. The measures after the cluster confirmed the diagnosis. Press reports in April 2024 said regulators had told some large brokers to halt any rise in their net exposure to over-the-counter derivatives linked to A shares, snowballs included, and on October 25, 2024 the Securities Association of China issued

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measures on income certificates under which a floating-income certificate's loss on its underlying may not exceed its fixed-income amount.

35.8 WORST-OF UNDERLYINGS

Section 34.11 priced the mechanism; here is what it does commercially. Most autocallables reference the *worst* of two to four underlyings. This raises the coupon by far more than one extra underlying sounds worth, and Example 34.9 showed why: a worst-of put on three assets is worth 2.05 times a single-asset put at a correlation of 0.20, and still 1.49 times at 0.80, the level typical between major indices. That is why most notes are built this way: the coupon is what sells the note, and adding underlyings raises it without touching the barrier or the term, while "three major indices" sounds like diversification.

The disclosure problem is acute and simple: such a note is the opposite of diversified. The investor is exposed to whichever of the three performs worst, so adding an underlying can only increase the risk and can only increase the coupon.

The coupon's size is the signal. A coupon far above what a single-underlying note pays is not a better deal; it is a measure of how much extra risk was added, and it is the one number in the term sheet that reliably signals the risk the structure was designed to obscure.

35.9 THE ISSUER'S CREDIT

A structured note is an *unsecured obligation of the issuer*. The derivative is embedded in a bond, and if the bond defaults the derivative goes with it.

Principal protection means protection from the market, not from the issuer. That distinction was academic until September 2008, when retail holders of notes issued by a failed investment bank discovered that "100% principal protected" ranked alongside every other senior unsecured claim.

The pricing consequence is Example 35.1's uncomfortable corollary. The issuer's funding spread *is* the source of the note's attractive terms, so, other terms equal, a better-looking note is a worse credit. The investor is being paid for taking issuer credit risk, and in almost no case is that presented as the reason the terms are good.

Two developments have improved this and neither removes it. Disclosure rules now require the issuer's credit to be stated, and some jurisdictions require an **estimated initial value** (the note's fair value net of fees) to be printed on the term sheet. The estimated-value disclosure is the single most effective reform in this chapter's subject, because it makes Example 35.7's fee arithmetic visible without requiring the buyer to do it.

35.10 WHERE THE FEES ARE

The economics of the business are worth stating plainly, because they explain the product design. A structured note carries no explicit management fee. The compensation is the difference between the note's issue price and its fair value, typically 1.5% to 3% of face for a simple structure and more for a complex one, earned at issue.

This creates a specific incentive: complexity is profitable because it is hard to price. A buyer can check a participation rate against a competitor's; the fair value of a worst-of autocallable on three underlyings with a monthly autocall and a daily knock-in is not something a buyer can check at all. The margin is largest where the price is hardest to verify, which is the same structure Section 33.13 identified in non-standard swaps and Section 34.13 in exotics. The fee is not an estimate once the fair coupon is known, and Example 35.7 extracts it from the numbers already computed.

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Example 35.7 Reading the fee out of the coupon

Example 35.2's note prices at par on a coupon of **18.42%**. Suppose the same structure is issued at 1,000 paying **12.65%**.

The factor that converts a coupon into value is not the 0.70-year average life of Section 35.3. A note that knocks in and is never called runs the full two years and pays no coupon at all, and those paths, 16.45% of the total at two years each, account for 0.33 of the 0.70 years. The coupon-paying paths contribute the other 0.37, and discounting brings that to the factor the simulation gives directly: each point of annual coupon is worth **0.366** points of face. So the value given up is

$$(18.42\% - 12.65\%) \times 0.366 = 2.11\% \text{ of face}$$

and the note's fair value at issue is about **979** against an issue price of 1,000. The $21 is the fee, and it sits inside the coupon rather than beside it.

Three things follow that a term sheet does not say. *The fee is paid at issue and cannot be recovered by holding to maturity*: it is not an annual charge of 2.11% but a single 2.11% taken on day one, which on a note with a 0.70-year expected life is an annualized **3.0%**. *The buyer who sells early pays it twice*, once at issue and again in Section 35.4's exit spread. And *the fee is invisible in every scenario table*, because those tables are computed from the stated coupon and the stated coupon is already net of it.

This is precisely what the estimated-initial-value disclosure repairs. Printing "estimated initial value: $979 per $1,000" converts an argument about fairness into a number a buyer can compare across issuers, without requiring anyone to understand a single thing about barrier options. It does not attempt to explain the product; it prices it.

Illustrative magnitudes follow, scaled to complexity:

	Fair value	Gap to par
Principal-protected note, one index	0.9800	2.00%
Reverse convertible, one index	0.9750	2.50%
Autocallable, one index	0.9700	3.00%
Autocallable, worst-of three	0.9550	**4.50%**

The gap widens exactly as the buyer's ability to check it narrows. The defense is that the alternative is worse for the buyer. These structures cannot be assembled retail: the components are OTC, the sizes are institutional, and the hedging is continuous. A margin for manufacturing a payoff that could not otherwise be bought is a legitimate charge. The question is its size and whether it is visible, not whether it should exist.

35.11 HOW TO EVALUATE A NOTE

The chapter's material assembles into a procedure. It is short, it requires no model the reader does not now have, and it answers the questions a term sheet is designed not to raise. It has five steps:

1. *Decompose it.* Every note is a bond plus one or more options. Identify the bond, identify each option, and, critically, identify which side of each option the buyer is on. In Example 35.1 the buyer is long a call; in Section 35.2's reverse convertible and Example 35.2 the buyer is short a put. If the headline term is attractive, the buyer is short something.
2. *Price the bond leg at the issuer's own curve.* Example 35.1 showed that an attractive participation rate comes mostly from the issuer's funding spread: an issuer that pays more to borrow spends less on the zero-coupon bond and has more left to buy options. A note from an AA issuer and one from a BBB issuer are not comparable on their headline terms, and the difference between them is credit compensation, not investment skill.

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3. *Value the option leg and compare with par.* The gap is the fee. Where an estimated initial value is disclosed, this step is done for the reader; where it is not, the table in Section 35.10 gives the order of magnitude, and the rule that it widens with complexity is reliable.
4. *Simulate the distribution, not the scenarios.* A term sheet presents three or four illustrative outcomes. Example 35.3 shows what the term sheet omits: the skewness, the conditional loss given a loss, and the fact that the median and the 25th percentile can be the same number. A buyer who sees only the coupon and $P(\text{positive})$ has seen the two statistics the product was engineered to optimize.
5. *Ask what the exit is worth.* As Section 35.4 showed, the issuer is the only bid, the spread applies again, and the bid is lowest when the need to sell is greatest. A note should be evaluated as a hold-to-maturity instrument, because that is what it is.

Applied to Example 35.2's note, the five steps produce a description no marketing document contains: a two-year unsecured loan to the issuer, bundled with a written down-and-in put on an index, sold, at Example 35.7's 12.65% coupon, 2.1% of face above fair value, returning a positive amount with 84% probability (most often 3.2%, on average 5.6%, Example 35.3's figures at the lower coupon) and on average −32% with 16% probability, and not sellable except back to the issuer. That description is neither hostile nor favorable. It is the instrument, and a buyer who wants it after reading it is making an informed choice—which is the entire objective.

35.12 SUITABILITY, AND WHAT DISCLOSURE CANNOT FIX

Disclosure is necessary and it is not sufficient, and this chapter's material shows precisely where it stops working.

What disclosure fixes well. Fees and fair value: the estimated-initial-value requirement converts an invisible cost into a printed number, and it works because the quantity is a single figure the issuer already computes. Section 35.10's gap is exactly what disclosure is good at.

What it fixes poorly. A distribution cannot be disclosed into intuition. A buyer told that the product has a skewness of −1.59 has been given a true fact and no understanding, and a buyer shown three illustrative scenarios has been given an understanding that is wrong. The gap between Example 35.3's two columns is not a disclosure failure; it is a comprehension problem, and jurisdictions that have tried to solve it with more disclosure have generally produced longer documents and unchanged outcomes.

What it cannot fix at all is the correlation between the product's failure and everything else the buyer owns. An autocallable knocks in when equity markets fall, which is when the buyer's other assets have fallen, their employer is under pressure, and their appetite for risk has collapsed. No disclosure of the product in isolation reveals that, because the fact is about the portfolio and not about the note.

That points at the reform this chapter would actually endorse: not more disclosure, but a suitability test framed around the payoff's shape rather than the buyer's stated risk tolerance, because a buyer who says they want "income with capital preservation" has described a payoff the product does not have, and no confirmation from the buyer that they understand the risks changes that mismatch. Two limits on that recommendation need stating. Suitability regimes are paternalistic and impose real costs, including on buyers who understand the product perfectly well. And the demand is genuine: the yield problem these products solve is real, and a regime that removes them without addressing that demand pushes it somewhere less regulated. This is a trade-off rather than a solved problem, which is why Section 35.13 names the buyers rather than dismissing them.

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35.13 WHO BUYS THESE, AND WHY

Retail and private-bank clients supply the volume, and the demand is genuine rather than manufactured. The dominant motive is yield in a world where deposits pay less than the buyer needs, and the second is a preference for defined outcomes over open-ended ones. Both are real preferences and neither is irrational.

Insurers and pension funds buy principal-protected structures for the reason Section 31.17 gave: their liabilities contain guarantees, and a note with a floor matches them better than an unhedged equity holding.

Corporates occasionally buy them as treasury investments, and the suitability question is sharper there than in retail, because the sums are larger and the governance is often thinner.

Issuers sell them for funding and for the margin. The funding motive is underrated: a structured note is term funding raised from a source that does not otherwise lend to banks, and it is stickier than wholesale funding because the buyer cannot easily exit.

Dealers, frequently a different institution from the issuer, take the hedging risk, which is where Section 35.6's flows originate.

The recurring conclusion of Chapters 33 and 34 applies here in its sharpest form. The client is short the optionality and the dealer is long it, and here the client is a household. That does not make the trade illegitimate, but it does mean the disclosure question is not a formality: Section 35.9's estimated-value requirement answers its fee half, and Section 35.12 says why the distribution half needs more than disclosure.

SUMMARY

Every structured note decomposes into a bond and a derivative, and doing the decomposition converts a marketing document into a portfolio. A principal-protected note spends most of its face on the zero-coupon bond and buys participation with the rest, and the distributor's fee cuts that participation noticeably. Participation rises steeply with the issuer's funding rate while the index is unchanged, so the best-looking note on offer is systematically the weakest credit. A reverse convertible is a bond plus a short put that pays only below its barrier: its coupon above deposit rates is payment for a written put, described in fixed-income vocabulary while its market risk is entirely equity.

An autocallable with a daily knock-in prices at par on a coupon far above deposit rates. It pays its advertised return most of the time and loses in the rest, with small frequent gains set against rare large losses, so its returns are negatively skewed where the index's are positively skewed. A buyer moving from the index into the note exchanges a positively skewed exposure for a negatively skewed one with both earning the risk-free rate, and a variance-based risk measure reports the note as the safer of the two. It is also short volatility at the worst time, since volatility rises as the knock-in is approached.

Korea's retail market showed what this does at scale: the usual outcome repeated until it looked like a rule, and then a single vintage's barriers, maturities, and losses arrived together. A loss probability sold to a hundred thousand people in one quarter does not produce a matching fraction of them losing independently; it produces nearly all or none.

The issuer is long the embedded down-and-in put, whose delta grows past one in magnitude as the index falls toward the barrier and then jumps at it. At the barrier the issuer must sell at once and its stabilizing bid weakens, and because barriers cluster at round fractions of the level at issue, both happen together across issuers who each see only their own book. Worst-of underlyings raise the coupon substantially, and a note "linked to three major indices" is the opposite of diversified.

A note is unsecured issuer debt, so principal protection is protection from the market, not from the issuer. The fee is largest where the price is hardest to verify; disclosure fixes fees well

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and distributions poorly. The secondary market is the issuer alone, so the exit is worst when it is most wanted. The reform this chapter would endorse is a suitability test framed around the payoff's shape rather than the buyer's stated risk tolerance.

FURTHER READING

The regulatory literature is where this subject is treated, because supervisors have had to define what a fair disclosure looks like: the estimated-value rules and the key information document of the European Union's Regulation on Packaged Retail and Insurance-based Investment Products (PRIIPs, No. 1286/2014) are worth reading as primary sources, and their differences across jurisdictions are informative about what each regulator thought the problem was. On the academic side, the work measuring issue prices against model fair values (Henderson and Pearson; Célérier and Vallée) is consistent enough across markets and decades to be treated as settled: notes are issued at a premium to fair value, and the premium rises with complexity. Bergstresser and others on the retail structured-products market document the demand side. On autocallable hedging flows the public literature is thin and the useful material is in central-bank financial stability reports, which have covered the topic increasingly since 2022. For the pricing machinery, the references of Chapter 34 apply unchanged, since an autocallable is a barrier product with an early-termination feature; Section 21.11 adds what the forward smile does to it.

PRACTICE QUESTIONS

35.1 Using Example 35.1, compute the participation rate before and after a 2% fee, and explain why the same note's participation would be 59.4% at 2% issuer funding and 161.9% at 6%.

35.2 Explain why the best-looking note in a comparison across issuers at one moment is systematically the weakest credit.

35.3 Decompose a reverse convertible paying 10% when deposits pay 4%, and state what the 6% actually buys.

35.4 Using Example 35.2, state the fair coupon and the three probabilities, and explain why a payoff sold on its hit rate is negatively skewed.

35.5 Explain why an autocallable is short volatility in the worst possible way, being specific about when volatility rises.

35.6 State the note's and the index's skewness from the simulated distribution, and explain what a buyer switching between them is exchanging when both earn the risk-free rate.

35.7 State the average win, the average loss, and the worst one percent, and reconcile them with the 83.55% probability of a positive return.

35.8 Explain why the 25th and 50th percentiles of the note's return coincide, and why that makes the product unusually persuasive.

35.9 Explain why a variance-based risk measure reports this note as safer than the index, and what that says about variance as a risk measure.

35.10 Explain why the exit from a structured note is worst precisely when it is most wanted.

35.11 Explain why a product with a 17% loss probability sold to a hundred thousand people does not produce 17% of them losing independently.

35.12 Using Example 35.6, state the delta just above and just below the barrier and the jump, then compare the forced sale at the barrier with the buying the issuer loses below it.

35.13 Explain why barriers cluster across issuers, why no issuer can observe the clustering, and

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what that implies about what should be monitored.

35.14 Explain why a note linked to the worst of three indices pays a much higher coupon than a note on one of them, and why that is a measure of risk rather than of value.

35.15 List the five evaluation steps of Section 35.11, state what each is designed to reveal, and apply them to Example 35.2's note in one sentence.

FURTHER QUESTIONS

35.16 Example 35.1 shows participation nearly tripling with the issuer's funding rate. Derive the participation rate as a function of funding, maturity, and volatility; determine which of the three a buyer can most easily verify; and design a comparison metric that would rank notes across issuers on a like-for-like basis.

35.17 Example 35.2 finds a fair coupon of 18.42% with a 16.45% probability of loss. Compute the distribution of the investor's return rather than its mean, state its skewness, and use it to argue either that the product is or is not suitable for a buyer whose stated objective is income with capital preservation.

35.18 Example 35.6 shows the hedge delta jumping by 0.19 at the barrier. Aggregate the flow: assume a market-wide notional and a barrier distribution, derive the total hedging demand as a function of the index level, and identify the level at which the withdrawal of the bid is largest.

35.19 Section 35.7 argues that a stabilizer which works until it stops is not a stabilizer. Assess the claim rigorously: construct the case that autocallable hedging reduces realized volatility on average, construct the case that it redistributes volatility into the tail, and state what data would settle which effect dominates.

35.20 Section 35.10 argues the margin is largest where the price is hardest to verify. Test this across the products in Chapters 33–35: rank them by verifiability and by typical margin, and state whether the relationship holds and what explains any exception.

35.21 Section 35.4 argues the exit is worst when most wanted. Quantify it: model the issuer's bid as fair value less a spread that widens with volatility, and compute the loss an investor takes by exiting after a 20% fall relative to holding to maturity.

35.22 Section 35.12 argues that a suitability test should be framed around payoff shape rather than stated risk tolerance. Design such a test: specify the statistics it would require, the thresholds it would apply, and how it would classify each product in this chapter. Then identify the product it misclassifies and say why.

35.23 The estimated-initial-value disclosure tells a buyer what the note is worth on the day it is issued. Determine what it does *not* tell them, propose one further disclosure that would close the largest remaining gap, and explain why an issuer would resist it.

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